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DENVER EMU EAGLE P3 DOOR SYSTEM

Project # 7230

Door System

PTU Application
Software User Manual (SUM)

Contract Number	--	CDRL Number	--
		Customer Ref. Number	--

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Preface

This Software User Manual (SUM) complies with the requirements of the IEEE Std 1558-2004, IEEE J-Std 016-1995, IEEE Std 1063-2001 as well as with Rotem's, Supplier Software Specification.

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1 SCOPE

1.1 Identification

This document is the Software User Manual (SUM) for the Portable Test Unit (PTU) Application. The document number, revision, and date of this SUM are 201111466, Revision 1, and 2011-10-27 respectively. The software to which this document applies is the PTU Application with part number 2011L0054 for the DENVER EMU EAGLE P3 DOOR SYSTEM.

1.2 System Overview

The PTU Application software is used with the Door Control Unit (DCU) which is part of the Door System supplied by Vapor Stone Rail Systems (VSRS) to Rotem Hyunday for the DENVER EMU EAGLE P3 DOOR SYSTEM project. The Door System is installed on the new cars.

The PTU Application communicates with the DCU to execute user requests. Using the PTU Application, a user can maintain, diagnose and monitor the DCU. An overview of these functions is provided in Section 3.4 of this document. Section 3.3 provides the system requirements for running the PTU Application.

The intended audience of this Software User Manual (SUM) is the user of the Portable Test Unit (PTU) for the DENVER EMU EAGLE P3 DOOR SYSTEM Door Control Unit. The primary users of the PTU Application will be the DENVER EMU EAGLE P3 DOOR SYSTEM maintenance staff and engineering personnel troubleshooting and maintaining the Door System.

1.3 Document Overview

The purpose of this Software User Manual (SUM) is to describe how to use the PTU Application connected to a DCU for performing certain maintenance, diagnostic and monitoring functions.

The SUM includes the following:
Section 1 contains the identification, system and document overviews;
Section 2 lists all documents referenced;
Section 3 gives the summary of the software;
Section 4 provides procedures oriented to the first time user including software installation;
Section 5 is the Processing Reference Guide;
Section 5.3 gives the instructions on using the various functions of the PTU Application;
Section 5.7 provides tips on troubleshooting;
Section 6 lists the used acronyms;
Section 7 provides the document Index.

These sections help the user to have a better understanding of the PTU Application and its use.

2 REFERENCES

This section lists the number, title, revision, date, and source of all documents referenced in this manual.

Standards

IEEE Std 1063-2001	IEEE Standard for Software User Documentation, 2001-12-20, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems, 2005-03-23, IEEE, www.ieee.org
IEEE J-Std 016-1995	IEEE Trial-Use Standard for Information Technology Software Life Cycle Processes Software Development - Acquirer-Supplier Agreement, 1995-09-30, IEEE, www.ieee.org

Vapor Stone Rail Systems Documents

201111186	Software Configuration Management Plan (SCMP)	Rev. 2
201111207	Software Project Management Plan (SPMP)	Rev.00
201111208	Software Quality Assurance Plan (SQAP)	Rev.01
201111462	Software Requirements Specifications (SRS), PTU Application	Rev.01
201111185	System Functional Description (SFD)	Rev.04
201111489	Interface Control Document (ICD), Portable Test Interface Protocol	Rev.01

3 SOFTWARE SUMMARY

3.1 Software Application

The PTU Application software is designed for the Door Control Unit to help the maintenance staff in maintaining, inspecting, troubleshooting, adjusting, testing, and repairing. It is installed on an IBM compatible Laptop and runs under the Microsoft Windows Operating System.

3.2 Software Inventory

The PTU Application Software runs under Microsoft Windows.
NPlot plotting library is used for displaying door movement graphs.
DevExpress is used to generate reports.
This PTU Application SUM document in PDF format is used as the help file.

All required libraries, drivers, and help file are provided in the installation package.

Microsoft .NET Framework 3.5 (or higher) must be retrieved from the Microsoft website and installed in order for the PTU Application installer to complete successfully.

3.3 Software Environment

The PTU software is installed on a Windows-based portable computer.

Before proceeding with the installation of the PTU software, make sure the portable computer meets the following requirements:

- Computer type: IBM PC compatible;
- Operating system: Microsoft Windows XP (English), Windows Vista, and Windows 7; the PTU software is WIN32 API compatible;
- Processor: 3 GHz or faster;
- Memory: 1 GB base memory or more;
- Disks:
    - Drive "C", 160 GB HD or more;
    - CD-ROM Drive;
- Display mode: SVGA Color Display (minimum 1024x768);
- Serial Port:
    - Standard RS232 port;
- Other features:
    - One MS compatible mouse, track ball or equivalent;

To install the PTU software onto a PC, VSRS part number 2011L0054 is required; the part number represents a zipped file. The zipped file needs to be unzipped into a folder on the hard drive of the PC, and the file therein called Setup.exe needs to be run to initiate the installation process; details about the installation process are provided in Section 4.1.3 of this document. All the files needed for the proper installation of the PTU Application are contained within the above mentioned zipped file.

NOTE: Depending on the restrictions imposed by the group that administers PCs within the organization, only certain authorized personnel may be able to install/uninstall the PTU Application on the laptop. Consult the administrators regarding the privileges to install/uninstall software on PCs. No VSRS supplied password is needed for installation of the PTU Application onto a PC.

The PTU Application software communicates with the DCU Application or the Boot Loader via a direct RS232 link to the DCU.

3.4 Software Organization and Overview of Operations

The PTU Application software interfaces with the DCU Application via RS232. The PTU Application software is able to retrieve the DCU log data, to do real time monitoring of the DCU, to change door parameters, and to perform DCU software upload. It provides testing, analysis, and troubleshooting of the Door Control Unit.

The PTU Application presents the following functionalities:

- System troubleshooting: Door Movement Information, DCU Status, Real-Time Data Monitoring
- Data retrieval and viewing: Event Log file, Snapshot files, and System Counters
- Self-test result
- DCU Application upgrade
- DCU configuration: open and close timings, obstruction retries, door open duration passenger for duration enable, obstruction reopen distance, delay before reclosing, delay before closing
- DCU Time display and update
- Security Protection (password)
- Files and PTU Properties

Administration functions such as DCU Application upgrade, Log and Snapshot erasure, DCU Time update, resetting of System Counters, and DCU configuration changes are permitted only with administration access by use of a password.

3.5 Contingencies and Alternate States and Modes of Operation

The PTU Application can be logged into the DCU Application or the DCU Boot Loader Application. The normal use of the software is when it is logged into the DCU Application. By logging into the DCU Boot Loader, the PTU Application can be used with limited functionality to upload the DCU Application, view the System SW Version, download the Memory Dump, and view the DCU Code Status. By using the Login Option of PTU Properties, this option can be selected (see Section 5.3.3).

3.6 Security and Privacy

Making unauthorized copies of the software or documents are prohibited.

3.7 Assistance and Problem Reporting

All problems found during the operation of the PTU Application, related either to the documentation or the software application, should be reported directly to Product Support at Vapor Stone Rail Systems (phone: 514-335-4200).

Upon reception of the user’s problem notification, Vapor Stone Rail Systems is responsible for creating and addressing all System Problem Reports (SPR) related to the PTU Application.

4 ACCESSING THE SOFTWARE

4.1 First-time User

4.1.1 Equipment Familiarization

The user interface is a graphical interface (SVGA minimum 1024 X 768), and is developed for a host computer compatible with IBM-PC standard under Microsoft Windows Operating System.

An arrow shaped cursor or pointer on the screen shows the pointing device's current position. By moving the pointing device, the cursor can be placed over toolbar function buttons, menu selection items, and buttons on the PTU Control Menu. A left-click after positioning the pointer will launch the corresponding function.

The interface operates using both the keyboard and a pointing device (a mouse or a trackball). The commands are entered by activating screen buttons, pull-down menus, or using keyboard shortcuts.

4.1.2 Access Control

The PTU Application has a password security feature which allows access to administration functions when the password is entered. There is only one password and it is fixed. Password cannot be added, deleted, or changed.

4.1.3 Installation and Setup

The following steps describe how to install the PTU Application on a computer:

1. Start PTU installation package Setup.exe;
2. Follow the installation prompt to install the PTU Application into the default directory (“C:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*”, where *x.y.z* is the version);
3. Verify if any error message appears during the installation;
4. Verify if the “Setup was completed Successfully!” message pops up at the end of the installation process;
5. After the installation is finished, verify if the PTU software is installed in “C:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*” and “PTUProgram.exe” file is found in this directory (*x.y.z* is the version).

NOTE: If the .NET Framework is missing from the computer, the PTU Application installer will try to locate the .NET framework installation files. If they are not present, they must be retrieved and installed from the Microsoft website. The PTU Application installer will not complete successfully without a valid Microsoft .NET Framework 3.5 (or higher) installed. A reboot may be necessary for the .NET Framework installation.

The .NET framework and a RS232 device driver are installed with the PTU Application. It may be required to reboot the computer for the installation to complete.

If the PTU Application is launched following a problem during the installation process, certain error windows may pop-up; Table 5-2 includes a list of possible errors that may result from such problems.

NOTE: Only authorized people will be able to install the PTU Application on the laptop. Those authorized people also have the responsibility to update all the PTU laptops to the latest revision.

To uninstall the PTU software, please follow these steps:

1. From the control panel window, select the icon Add/Remove Programs;

2. Select the PTU Application (DENVER PTU: VAPOR DOOR Version *x.y.z*", where *x.y.z* is the version) and confirm the desire to remove the application;

3. After the Removal is finished, verify that the directory:
   C:\DENVERPTU\Program\DOOR-Door Subsystem\DOOR-*x.y.z*
   is removed.

NOTE: Only authorized people will be able to uninstall the PTU Application from the laptop. Those authorized people also have the responsibility to update all the PTU laptops to the latest revision.

4.2 Initiating a Session

Before a session can be started with a DCU, make sure that the portable computer running the PTU software is directly connected to the DCU via the RS232 port .

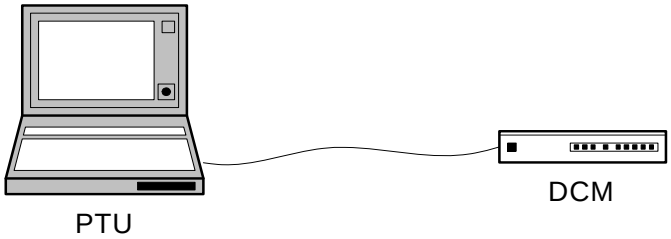


Figure 4-1: PTU Start-up

The start-up procedure for the PTU is the responsibility of Rotem. See Rotem's start-up procedure on how to start the PTU Application. If administration functions are required, the password must be entered after the PTU Application is launched. Since only authorized personnel will get administration access, the password is not provided with this manual.

Assuming that there were no errors in the installation process, the error that is most likely to be observed is an error in the RS232 connection. In such a case, the grayed out disconnected icon of Figure 4-2 will appear in the PTU Status bar. If this occurs, verify that the RS232 cable is connected properly.

Figure 4-2: Indicator for Missing RS232 Connection

4.3 Stopping or Suspending the Software

To cease using the software, perform the following steps:

1. Click *File* → *Exit* from the PTU menu or click on Exit from the PTU Control Menu.

2. A window will open asking to save the project file. Click on Yes if the project file is to be saved.

3. The PTU terminates itself and the PTU main window is closed.

5 PROCESSING REFERENCE GUIDE

5.1 Capabilities

The functions of the PTU Application can be accessed through the PTU main menu, PTU Control Menu, or the toolbar. All the functions available on the PTU Control Menu are also available from the PTU main menu. The toolbar contains buttons for some functions available on the PTU main menu as well as some available on both the PTU main menu and the PTU Control Menu.

There is one function on the toolbar that can only be accessed from the toolbar and is not available from any other menu. This functions is the Stop Function used to cancel the current file download or upload.

The PTU Control Menu contains only the functions used for maintenance and administration of the DCU.

For more information on the PTU Application's functionalities, see Section 3.4.

5.2 Conventions

Events in the Event Log that have a state of "Set" and Self Test failure results are displayed in red.

Abbreviated vocabulary is used in the PTU Application and their full representation can be found in the Acronyms Section 6.2.

Other conventions are derived from requirements described in the PTU SRS document or client specification.

5.3 Processing Procedure

Sections 5.3.1 PTU Application Main and 5.3.2 PTU Control Menu provide information on how to access the various features of the PTU Application. The subsequent subsections provide information on the actual use of these features of the PTU Application.

5.3.1 PTU Application Main

5.3.1.1 Scope

The following subsections provide information regarding the features that a user can access through the PTU Application's main window.

5.3.1.2 Method

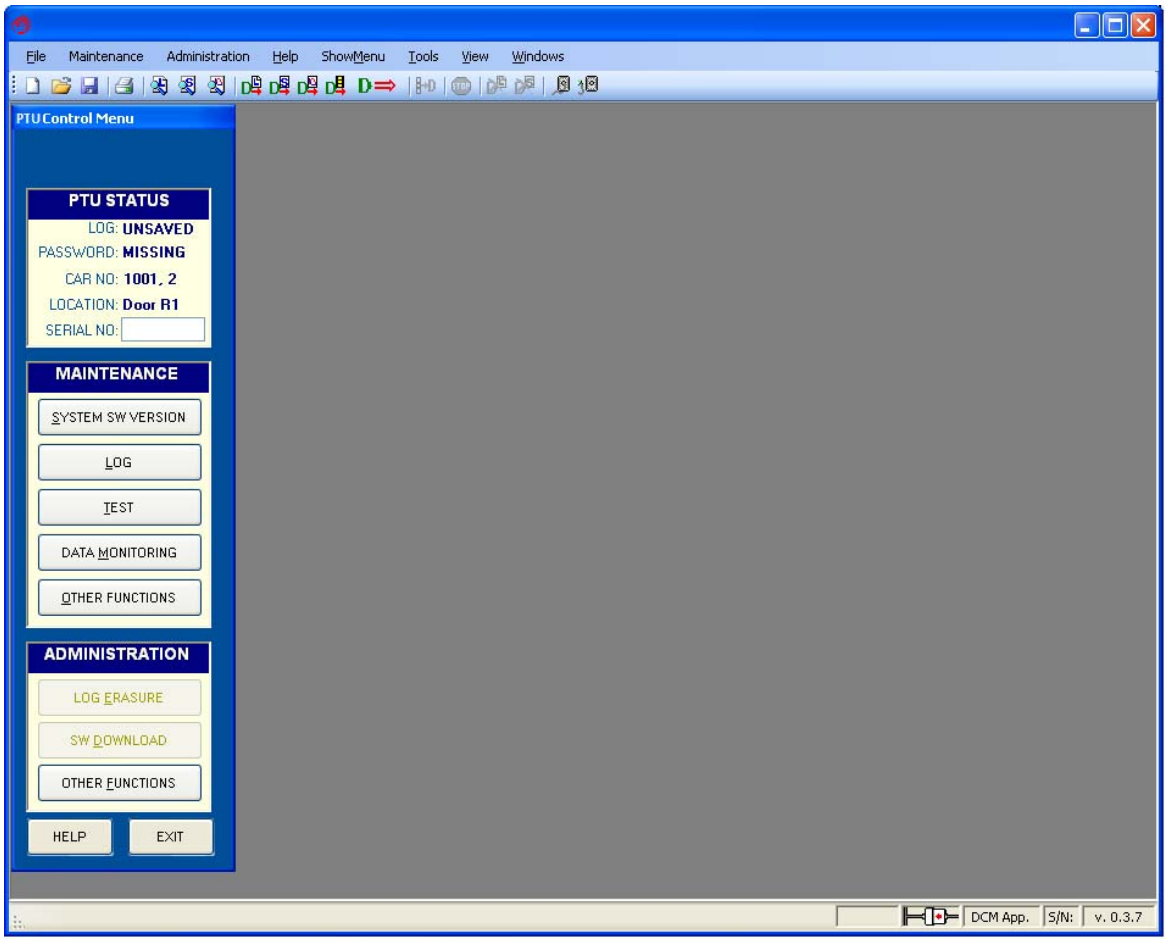


Figure 5-1: PTU Application Main User Interface

The PTU Application uses the *Multiple-Document Interface* (*MDI*). As shown in Figure 5-1, the Main Window includes Menu, Toolbar, Status bar, and PTU Control Menu.

The Menu presents all operations of PTU Application. The details are as follow:

- **File**
  - New Project: Create a new PTU project.
  - Open Project: Open a PTU data file.
  - Save Project: Save the current PTU project.
  - Save Project As: Save the current PTU project to user selected file.
  - Save Log As: Save the current Log to a text file.

- Properties: Open the PTU properties window.
- Print: Print the current PTU project, including File Summary, Event Log, and Snapshots.
- Print OPDF: Print the Operating Parameters Description File.
- Print EDF: Print the Event Description File.
- Exit: Exit the PTU application.

- **Maintenance**
  - System SW Version: Display the DCU software versions.
  - Log : Open the PTU project event log window.
  - Test: Display the DCU Self-Test.
  - Data Monitoring: Monitor the DCU Status.
  - Other Functions
    - Snapshot View: Open the PTU project snapshot list window.
    - Snapshot Download: Download the current DCU snapshots.
    - Log Download: Download the current DCU event log.
    - Graph View: Open the PTU Graph View window.
    - Graph Download: Download and display the Door Graph.
    - Memory Dump: Open the Memory Dump window.
    - Browser: Perform real time monitoring of all DCU operating parameters.
    - System Counters View: Open the System Counters window.
    - System Counters Download: Download the System Counters information.
    - Download All: Download the event log, snapshots, door graphs, and system counters.
- **Administration**
  - Log Erasure: Delete the DCU event log.
  - SW Download: Open the SW Download window.
  - Other Functions
    - Load Flash (for Boot Loader Upgrade)
      - Open Flash File: Open Boot Loader file.
      - Send Flash File: Send Boot Loader file to DCU.
    - NodeUtil (for Neuron Upgrade): Open NodeUtil software tool to download.
    - DCU Configure: Open the DCU Config window.
    - Snapshot Erasure: Open the Snapshot Erasure window.
    - Password: Enter the password to unlock administration function.
- **Help**
  - Help Contents: Open the PTU help file (PTU Application Software User Manual).
  - About: Open the PTU About window.
  - Web Home: Open the Vapor Stone Rail Systems web page with Internet explorer.
- **ShowMenu**: Set the focus on the PTU Control Menu.
- **Tools**
  - DCU Time: Open the DCU Time window.
  - Operating Parameters List: Open the Operating Parameters List.
  - Event Description List: Open the Event Description list.
  - Update EDF and OPDF: Update the EDF (Event Description File) and OPDF (Operating Parameters Description File) residing on PTU computer from the current DCU.
  - DCU Code Status: Retrieve and display the current DCU Boot Loader and DCU application's size, CRC information.
  - Debug
    - Communication Debug: Open Communication Debug window to view messages sent to and received from the DCU.
    - Error Log: Open PTU Application Error Log to view error messages.
  - Login: Login to the connected DCU.
  - Logout: Logout from the connected DCU.
- **View**
  - Toolbar: Hide/Display the PTU Toolbar.
  - Status bar: Hide/Display the PTU Status bar.

- **Windows**
  - Cascade: Arrange the PTU open windows in Cascade mode.
  - Tile Horizontal: Arrange the PTU open windows in Tile Horizontal mode.
  - Tile Vertical: Arrange the PTU open windows in Tile Vertical mode.
  - Arrange Icons: Arrange the PTU open windows Icons.
  - Close All Windows: Close all the PTU open windows including the Control Menu.

The Toolbar links to the following functions:

- New: Same as the menu *File → New Project*.
- Open: Same as the menu *File → Open Project*.
- Save: Same as the menu *File → Save Project*.
- Print: Same as the menu *File → Print*.
- Log View: Same as the menu *Maintenance → Log*.
- Snapshot View: Same as the menu *Maintenance → Other Functions →Snapshot View*.
- Graph View: Same as the menu *Maintenance → Other Functions → Graph View*.
- Log Download: Same as the menu *Maintenance* → Other Functions → Log Download.
- Snapshot Download: Same as the menu *Maintenance → Other Functions → Snapshot Download*.
- Graph Download: Same as the menu *Maintenance → Other Functions → Graph Download*.
- Memory Dump Download: Download the DCU SRAM and NVRAM contents.
- Download All: Same as the menu *Maintenance → Other Functions →Download All*.
- SW Application Download: Same as the menu *Administration → SW Download*.
- Stop: Stop the current file download/upload (Upgrade).
- Log Erasure: Same as the menu *Administration → Log Erasure*.
- Snapshot Erasure: Same as the menu *Administration → Other Functions → Snapshot Erasure*.
- Login: Same as the menu *Tools → Login.*
- Logout: Same as the menu *Tools → Logout*.

The Status bar presents the following information:

- Communication Device: The indication of the PTU Serial Communication Port Number
- Communication State: The indication of PTU software login is represented by a connection icon. A connected icon indicates that the PTU is logged into the DCU.
- Communication Target: The indication of PTU current communicating DCU software: DCU App. or Boot Loader.
- PTU Application Version: The PTU software version.

5.3.1.3 Related Information

None

5.3.1.4 Cautions and Warnings

Whenever the PTU Application is launched with the DCU connected to the PTU, always verify that proper communication is established by ensuring the connection icon in the Status Bar. Otherwise, double check the status of the cable and ensure that the DCU is powered on.

5.3.2 PTU Control Menu

5.3.2.1 Scope

The following subsections provide information regarding the features that a user can access through the PTU Control Menu. It appears prominently in the main window of the PTU Application. All of these features can also be accessed through the application's drop-down menu at the top, and many of them can also be accessed through the application's toolbar.

5.3.2.2 Method

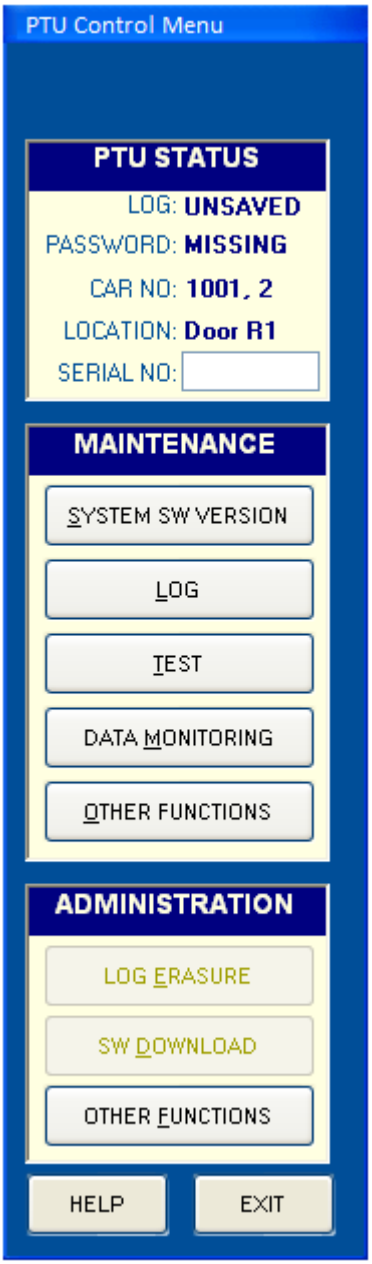


Figure 5-2: PTU Control Menu User Interface

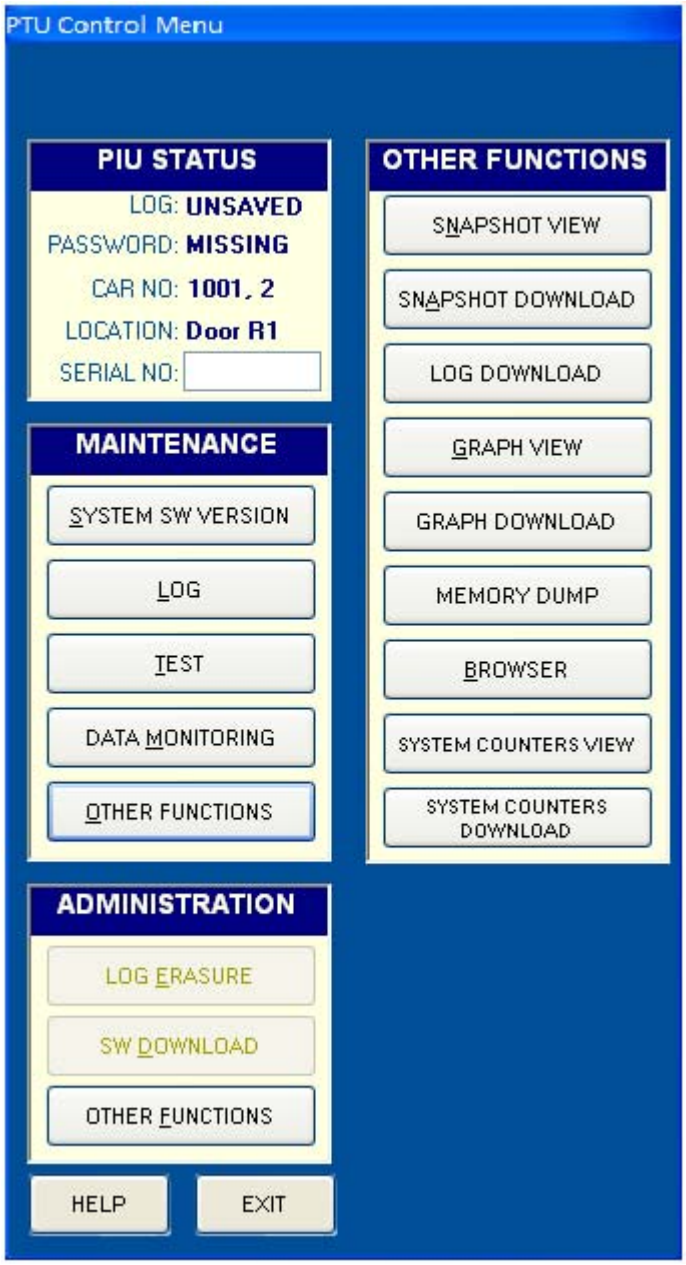


Figure 5-3: PTU Control Menu (Maintenance Other Functions) User Interface

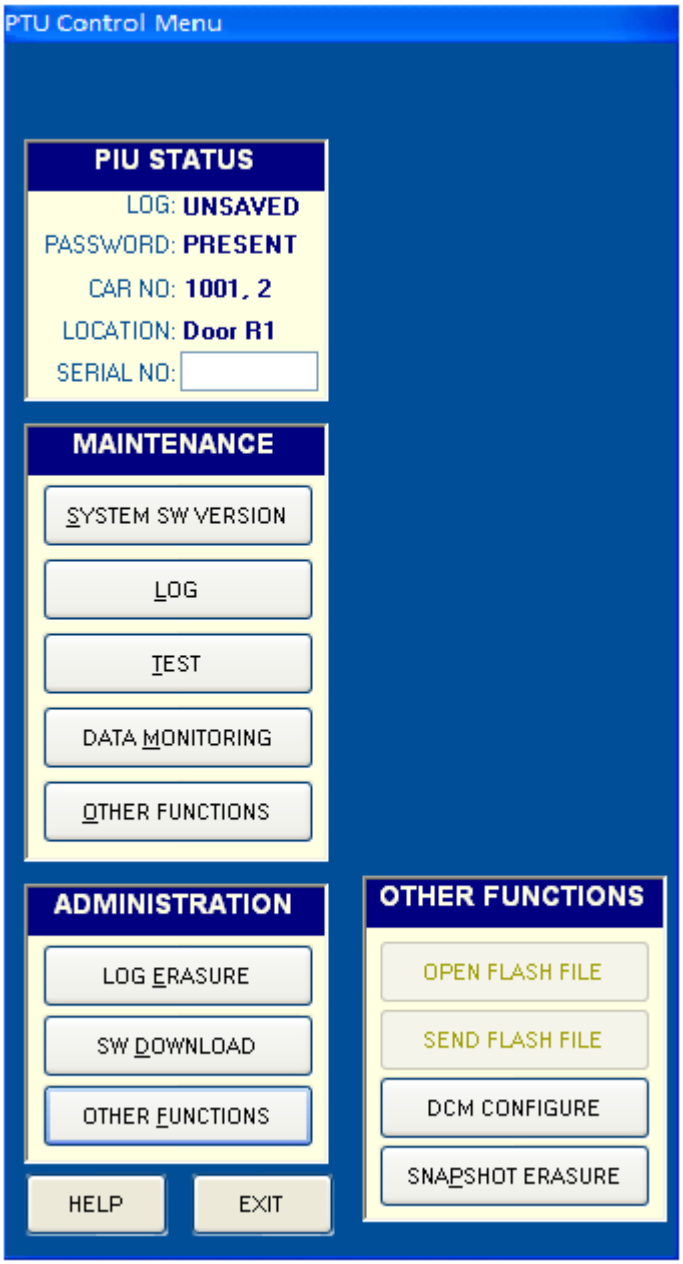


Figure 5-4: PTU Control Menu (Administration Other Functions) User Interface

The PTU Control Menu appears on the PTU main window when the PTU Application is started.

As shown in Figure 5-2, there are the Log Status (SAVED or UNSAVED), Password status (MISSING or PRESENT), Car Number with car position, Door Location information, and a set of command buttons. In the Serial Number box, the user is able to input the DCU Serial Number.

NOTE: The buttons in the Control Menu provide shortcuts to many options already included in the menu bar. To know the functions of these items, refer to Section 5.3.1.2.

As shown in Figure 5-2, Figure 5-3, and Figure 5-4, the command buttons are as follows:

- **MAINTENANCE**
  - SYSTEM SW VERSION
  - LOG
  - TEST
  - DATA MONITORING
  - OTHER FUNCTIONS
    - SNAPSHOT VIEW
    - SNAPSHOT DOWNLOAD
    - LOG DOWNLOAD
    - GRAPH VIEW
    - GRAPH DOWNLOAD
    - MEMORY DUMP
    - BROWSER
    - SYSTEM COUNTERS VIEW
    - SYSTEM COUNTERS DOWNLOAD
- **ADMINISTRATION**
  - LOG ERASURE
  - SW DOWNLOAD
  - OTHER FUNCTIONS
    - OPEN FLASH FILE
    - SEND FLASH FILE
    - DCU CONFIGURE
    - SNAPSHOT ERASURE
- **HELP**
- **EXIT**

5.3.2.3 Related Information

None

5.3.2.4 Cautions and Warnings

Always verify that proper communication is established between the DCU and the PTU Application by ensuring the connection icon in the Status Bar indicates connection with the DCU. Otherwise, double check the status of the cable and ensure that the DCU is powered on.

5.3.3 PTU Properties

5.3.3.1 Scope

The following subsections provide information on setting the basic communication properties of the PTU Application.

5.3.3.2 Method

To change the PTU properties, click *File → Properties* from the PTU menu.

As shown in Figure 5-5, the Properties window allows the user to modify the communication device and the PTU login option.

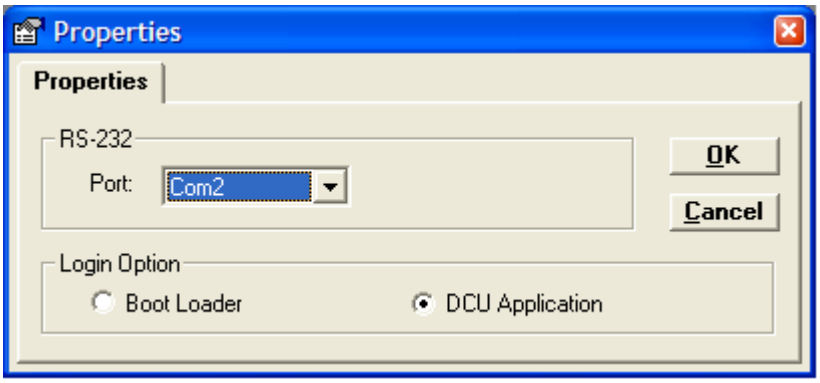


Figure 5-5: PTU Properties User Interface

RS-232

When the properties window is open, the current PTU communication port is displayed. The user is allowed to change the Port number if the PTU is not logged into any DCU.

Login Option

The Login Option allows the user to log into different applications (DCU Boot Loader or DCU Application) when the DCU is booted up. Normally, the PTU should log into the DCU Application and interface with the DCU Application to retrieve the DCU information and to troubleshoot. If the DCU Application does not work properly for some reason and cause the PTU communication to fail, the Login Option could allow the user to log into the DCU Boot Loader and to reload the DCU Application. Also, the user can use this option to log into the DCU Boot Loader to test all the DCU Boot Loader functions.

Two command buttons are used to save or cancel the user input selections:

- OK button: Apply and save the current user settings;
- Cancel button: Exit properties without any modification to the PTU settings.

5.3.3.3 Related Information

None

5.3.3.4 Cautions and Warnings

With the DCU Application working as intended, and under normal operating conditions, the Login Option should be set for DCU Application.

5.3.4 Login/Logout

5.3.4.1 Scope

The following subsections provide information on how to Login to and Logout of the DCU.

5.3.4.2 Method

Login to the DCU:

1. Click *Tools* → *Login* from the PTU menu, click the Login button on the toolbar or click the disconnected icon on the Status bar.

Figure 5-6: Status Bar – Disconnected Icon

Note: The version numbers in these screenshots are for example purposes only and may not reflect the latest software version.

2. Once connected, the connection icon on the Status bar will change to the connected icon. The communication target of either DCU App. or Boot Loader is also displayed on the Status bar (see Login Option in Section 5.3.3).

Figure 5-7: Status Bar – Connected Icon

Note: The version numbers in these screenshots are for example purposes only and may not reflect the latest software version.

Logout of the DCU:

1. Click *Tools* → *Logout* from the PTU menu, click the Logout button on the toolbar or click the connected icon on the Status bar.

5.3.4.3 Related Information

- Once the PTU Application is connected to the DCU, it will automatically request for the Location ID, EDF & OPDF versions, and software versions from the DCU. The Car Number with the Car Position and Door Location are then displayed in the PTU Status of the PTU Control Menu.

- If the EDF or OPDF version number returned by the DCU is greater than the ones loaded in the PTU Application, an automatic update of the EDF and OPDF will be done.

5.3.4.4 Cautions and Warnings

None

5.3.5 Password

5.3.5.1 Scope

The following subsections provide information on how to obtain access to administration functions by entering the administrator password.

5.3.5.2 Method

1. In the menu, click *Administration → Other Functions → Password* or click on the word "MISSING" in the Password field of PTU Status on the PTU Control Menu.

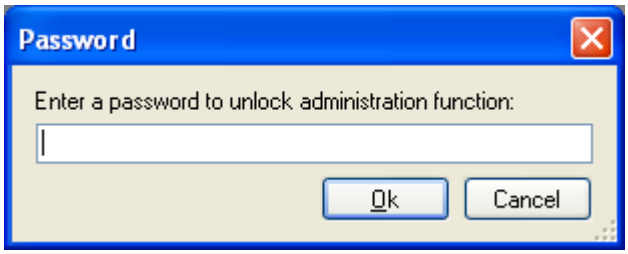


Figure 5-8: Password Entry User Interface

2. Enter the password in the text box provided by the Password window.

The Password window also provides command buttons:

- OK button: After the correct password is entered into the text box, clicking the OK button will unlock administration functions.
- Cancel button: Cancel the password entry and close the Password window.

Once the administration functions are unlocked, the Password field of the PTU Control Menu changes to "PRESENT".

The following administration functions are unlocked:

- Log Erasure
- Snapshot Erasure
- DCU Configuration changes
- DCU Application upgrade
- DCU Boot Loader upgrade
- DCU Time update
- System Counters Reset

5.3.5.3 Related Information

None

5.3.5.4 Cautions and Warnings

The PTU should not be left unattended after the password has been entered in order to prevent unauthorized access to the administration functions.

5.3.6 Test

5.3.6.1 Scope

The following subsections provide information on results of an internal Self-Test on the DCU.

5.3.6.2 Method

To request the DCU self test results, click *Maintenance* → *Test* or click on "Test" from the PTU Control Menu (Maintenance).

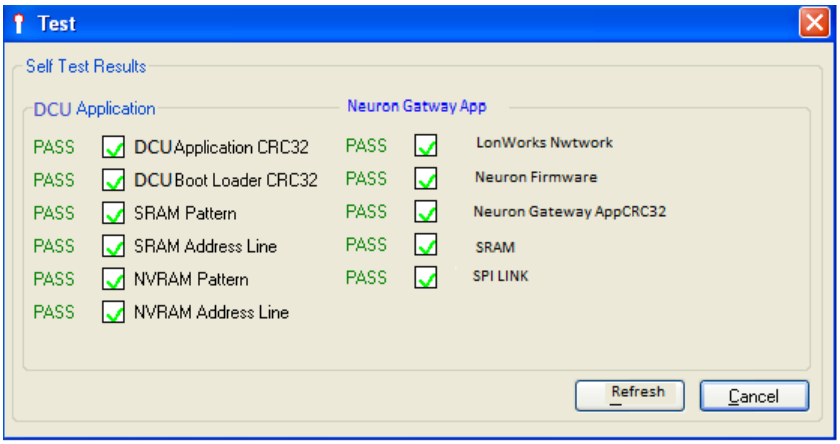


Figure 5-9: Test User Interface

The Test window shows current Self-test processing and results. The devices that are tested are listed in this window. If the device passes the Self-test, the word "PASS" and the symbol ✓ appear in the front of the device name; otherwise, the word "FAIL" in red and the symbol ✗ appear.

As shown in Figure 5-9, the following listed devices are tested:

- DCU Application
  - DCU Application CRC32
  - DCU Boot Loader CRC32
  - SRAM Pattern
  - SRAM Address Line
  - NVRAM Pattern
  - NVRAM Address Line
- Neuron Gateway Application
  - LonWorks Network
  - Neuron Firmware
  - Neuron Gateway App CRC32
  - SRAM
  - SPI Link

The Test window also provides command buttons:

- Refresh button: Refresh the Self-test Results;
- Cancel button: Exit the Test window.

5.3.6.3 Related Information

None

5.3.6.4 Cautions and Warnings

Failures in test items that relate to the SRAM or the NVRAM may require the DCU to be shipped back to VSRS. Failures related to CRC32 may require re-installation of the corresponding SCI (refer to Sections 5.3.25 and 5.3.26).

5.3.7 Data Monitoring

5.3.7.1 Scope

The following subsections provide information on how to monitor the status of the various sense points available to the DCU Application, and to monitor the currently active events within the DCU.

5.3.7.2 Method

The purpose of Status window is to give the user an overview of the DCU Status. The DCU Status signals are grouped into DCU Status, Power Supply, DCU Outputs, DCU Inputs, and Crew Key Switch. Each time the signal state is changed, it is shown on the related group window. The signals are represented by a simulated LED indicator with the On state being Green, Yellow, or Red. The color attributed to their On state is determined by the real color of the LED on the DCU board. The Off state is indicated by a Black LED-Off representation. The active event is also displayed on the event table. The Figure 5-10 shows the interface window.

Select the menu *Maintenance → Data Monitoring* or click on "Data Monitoring" from the PTU Control Menu (Maintenance) and the PTU will open the DCU Status window.

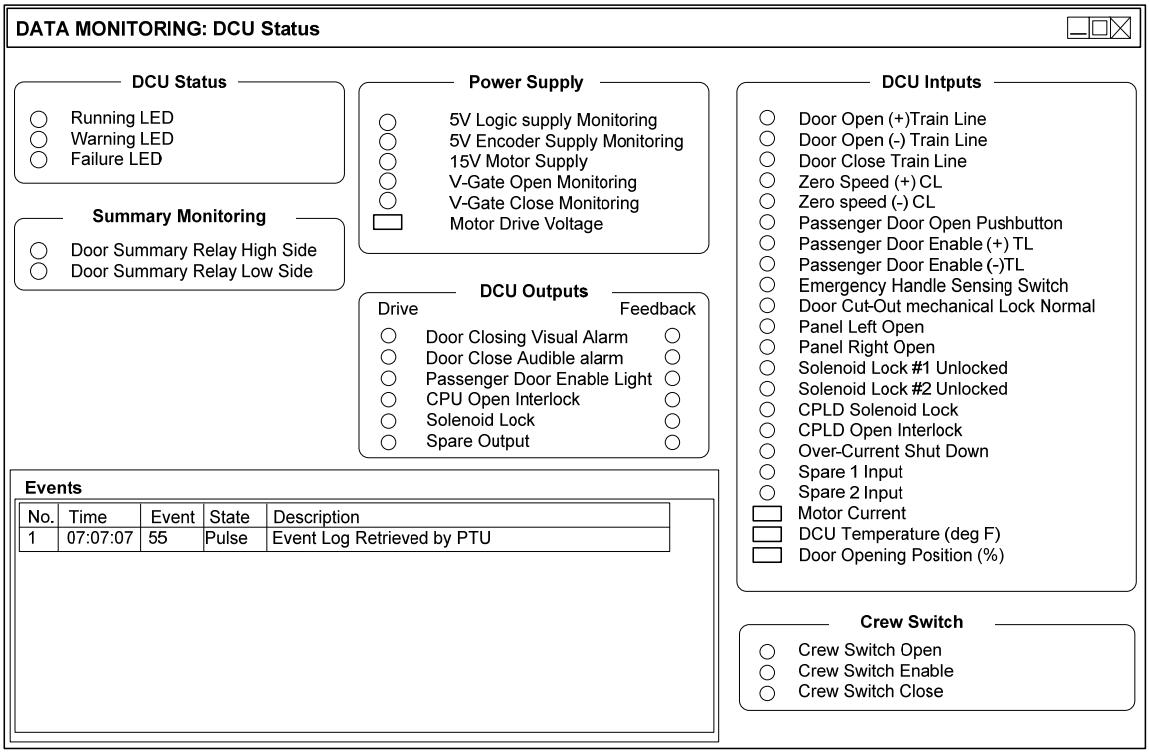


Figure 5-10: Data Monitoring User Interface[1]

[1] Note : Figure 5-10 is a preliminary interface window.

Signal Groups are described as follows:

Denver Door System Signal	Description
DCU Inputs	
Door Open (+) Train Line	On/Off signal. The LED goes green (On) while the Open (+) Train Line is energized.
Door Open (-) Train Line	On/Off signal. The LED goes green (On) while the Open (-) Train Line is energized.
Door Close Train Line	On/Off signal. The LED goes green (On) while the Close Train Line is energized.
Zero Speed (+) Car Line	On/Off signal. The LED goes green (On) while the Zero Speed (+) Car Line is energized.
Zero Speed (-) Car Line	On/Off signal. The LED goes green (On) while the Zero Speed (-) Car Line is energized.
Passenger door Open Pushbutton	On/Off signal. The LED goes (On) while Passenger Door Open Pushbutton is energized.
Passenger Door Enable (+) TL	On/Off signal. The LED goes green (On) while the Passenger Door Enable (+) Train Line is energized.
Passenger Door Enable (-) TL	On/Off signal. The LED goes green (On) while the Passenger Door Enable (-) Train Line is energized.
Emergency Handle Sensing Switch	On/Off signal. The LED goes (On) while the Emergency Handle Sensing Switch is actuated.
Door Cut-Out Mechanical Lock Normal	On/Off signal. The LED goes (On) while Door Cut-Out Mechanical Lock is in Normal.
Panel Left Open	On/Off signal. The LED goes green (On) while the Panel Sense switch left indicates door open.
Panel Right Open	On/Off signal. The LED goes green (On) while the Panel Sense switch right indicates door open.
Solenoid Lock #1 Unlocked	On/Off signal. The LED goes green (On) while the Solenoid Lock Sensing Switch #1 indicates unlocked.
Solenoid Lock #2 Unlocked	On/Off signal. The LED goes green (On) while the Solenoid Lock Sensing Switch #2 indicates unlocked.
CPLD Solenoid Lock	On/Off signal. The LED goes green (On) while the CPLD Solenoid Lock power is energized.
CPLD Open Interlock	On/Off signal. The LED goes green (On) while the CPLD Open Interlock power is energized.
Over-Current Shutdown	On/Off signal. The LED goes yellow (On) while the Motor Over Current is energized.
Spare 1 Input	On/Off signal. The LED goes green (On) while the Spare 1 is energized.
Spare 2 Input	On/Off signal. The LED goes green (On) while the Spare 2 is energized.
Motor Current	Indicates the motor current (A).
DCU Temperature	Indicates the internal DCU Temperature in Fahrenheit.
Door Opening Position (%)	Indicates the door position as a percentage of full open.
Crew Switch	
Crew Switch Open	On/Off signal. The LED goes green (On) while the switch is turned to open position.
Crew Switch Enable	On/Off signal. The LED goes green (On) while the switch is turned to open position.
Crew Switch Close	On/Off signal. The LED goes green (On) while the switch is turned to Close position.
DCU Outputs	
Door Closing Visual Alarm	On/Off signal. The LED goes (On) while the Door Closing

Denver Door System Signal	Description
	Visual Alarm is energized.
Door Closing Visual Alarm Feedback	On/Off signal. The LED goes green (On) while the Door Closing Visual Alarm Feedback is energized.
Door Closing Audible Alarm	On/Off signal. The LED goes (On) while the Door Closing Audible Alarm is energized.
Door Closing Audible Alarm Feedback	On/Off signal. The LED goes green (On) while the Door Closing Audible Alarm Feedback is energized.
Passenger Door Enable Light	On/Off signal. The LED goes (On) while the Passenger Door Enable Light is energized.
Passenger Door Enable Light Feedback	On/Off signal. The LED goes green (On) while the Passenger Door Enable Light Feedback is energized.
CPU Open Interlock	On/Off signal. The LED goes green (On) while the CPU Open Interlock is energized.
CPU Open Interlock Feedback	On/Off signal. The LED goes green (On) while the CPU Open Interlock Feedback is energized.
Solenoid Lock	On/Off signal. The LED goes green (On) while the Solenoid Lock power is energized.
Solenoid Lock Feedback	On/Off signal. The LED goes green (On) while the Solenoid Lock power Feedback is energized.
Spare Out	On/Off signal. The LED goes green (On) while Spare Out is energized.
Spare Out Feedback	On/Off signal. The LED goes green (On) while the Spare Out Feedback is energized.
Power Supply	
5V Logic Supply Monitoring	On/Off signal. The LED goes green (On) while the Logic Supply Monitoring is within normal range
5V Encoder Supply Monitoring	On/Off signal. The LED goes green (On) while Encoder Supply Monitoring is within normal range
15V Motor Supply Monitoring	On/Off signal. The LED goes green (On) while the Motor Supply Monitoring is within normal range
V-Gate Open Monitoring	On/Off signal. The LED goes green (On) while V-Gate Open Monitoring is within normal range
V-Gate Close Monitoring	On/Off signal. The LED goes green (On) while V-Gate Close Monitoring is within normal range
Motor Drive Voltage	Shows the Motor Voltage Reading Value
DCU Status	
Running LED	On/Off signal. The LED goes green (On) while the DCU is running.
Warning LED	On/Off signal. The LED goes yellow (On) while an event with Exception level is active.
Fault LED	On/Off signal. The LED goes red (On) while an event with Failure level is active.
Summary Monitoring	
Door Summary Relay High Side	On/Off signal. The LED goes green (On) while the Door Summary Relay High Side is energized
Door summary Relay Low Side	On/Off signal. The LED goes green (On) while the Door Summary Relay Low Side is energized.

The Events table has the following columns:

- No.: The sequence No. of the event in the Event table.
- Time: The PTU computer time when the event is received.
- Event: The event code
- State: The state of the event (Set/Reset or Reset)
- Description: The description of the event.

Right-clicking on the Events table will bring up a selection box with the following options:

- Sync. Current Event: Request the DCU to transmit the current fault that is in set state.
- Clear Event Table: Clears the display of events in the PTU window without affecting the status of events within the DCU.

5.3.7.3 Related Information

None

5.3.7.4 Cautions and Warnings

Once proper communication is established between the DCU Application and the PTU Application, there should be no problem monitoring the DCU Status in real time. Otherwise, double check the status of the RS232 cable and investigate other possible hardware problems.

5.3.8 Browser

5.3.8.1 Scope

The following subsections provide information on how to use the Browser window to browse all the available operating parameters for the DCU. These parameters are defined in the Operating Parameters Description File (OPDF).

5.3.8.2 Method

To perform real time monitoring of the DCU operating parameter, click menu *Maintenance → Other Functions → Browser* or click on “Other Functions” then “Browser” from the PTU Control Menu (Maintenance) to open the Browser window.

DATA MONITORING: BROWSER

Para.ID	Description	Unit	Value
1	Motor Current	A	0.0
2	Door Panel Opening Position	%	0
3	DCU Temperature	oF	71
4	Solenoid Lock Switch 1		1
5	Solenoid Lock Switch 2		1
6	Panel Sensing Switch Left		1
7	Panel Sensing Switch Right		1
8	Zero Speed PWR CL		1
9	Zero Speed RTN CL		1
10	Open PWR TL		1
11	Open RTN TL		1
12	Close TL		1
13	Crew Switch Open		1
14	Crew Switch Enable		1
15	Crew Switch Close		1

Figure 5-11: Browser User Interface[2]

As shown in Figure 5-11, the Browser table has the following columns:

- Para. ID: Parameter ID Number;
- Description: Parameter Description;
- Unit: Parameter Unit;
- Value: Value of the DCU Parameter (For data with only two states (On/Off), a light-bulb-on image is displayed beside the value 1 and a light-bulb-off image is displayed for a value of 0.

5.3.8.3 Related Information

None

5.3.8.4 Cautions and Warnings

If a DCU is disconnected, its status will stay at the last value received.

[2] List of signals in the snapshot is for example purpose only and may not reflect the Denver Project’s signals.

5.3.9 Memory Dump

5.3.9.1 Scope

The following subsections provide information on how to use the Memory Dump window to view and save the DCU SRAM (starting at 0xE000) and NVRAM (starting at 0x120000) memory contents. The memory contents are displayed in Hex and ASCII.

5.3.9.2 Method

To display the memory, click menu *Maintenance → Other Functions → Memory Dump* or click on "Other Functions" then "Memory Dump" on the PTU Control Menu (Maintenance) .

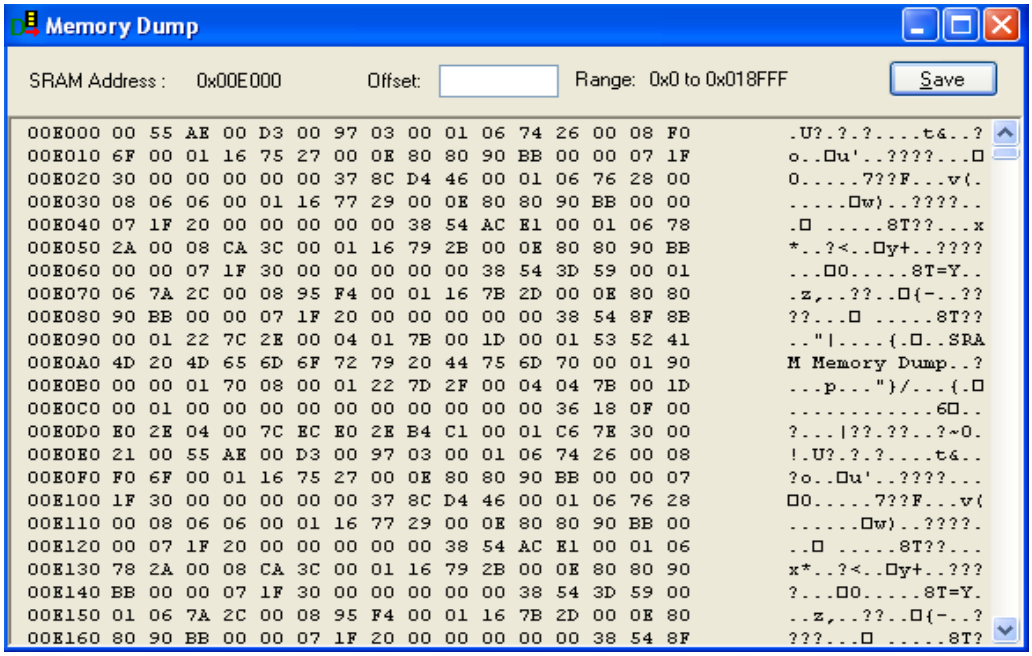


Figure 5-12: Memory Dump User Interface

As shown in Figure 5-12, the address and the memory values in both Hex and ASCII formats are presented. There are also the following text box and command button:

- Offset text box: The Offset text box can be used to search and display the memory content based on an offset from the starting Memory Dump contents;
- Save button: Save the memory contents to a binary file.

5.3.9.3 Related Information

None

5.3.9.4 Cautions and Warnings

The mere launching of the window will only display the memory contents that is currently in the PTU Application database. The Memory Dump Download button on the toolbar needs to be clicked to download and display the most current memory contents. Also, refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.10 Memory Dump Download

5.3.10.1 Scope

The following subsections provide information on how to download the latest DCU memory (SRAM and NVRAM) contents that are used to update the PTU Application's database for display in the Memory Dump window.

5.3.10.2 Method

Click the Memory Dump Download button on the toolbar.

A message window (Figure 5-13) pops up to show the download process when the PTU Application is performing the download from the DCU. The window is closed after the download is finished. The Memory Dump window (Figure 5-12) is then opened and the memory contents are displayed.

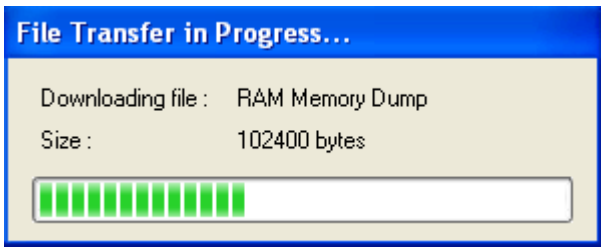


Figure 5-13: Memory Dump Download Progress Window

5.3.10.3 Related Information

None

5.3.10.4 Cautions and Warnings

Refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.11 System SW Version

5.3.11.1 Scope

The following subsections provide information on how to display the version information, including their compilation date, for all the SCIs in the DCU.

5.3.11.2 Method

Click *Maintenance → System SW Version* or click on "System SW Version" from the PTU Control Menu (Maintenance).

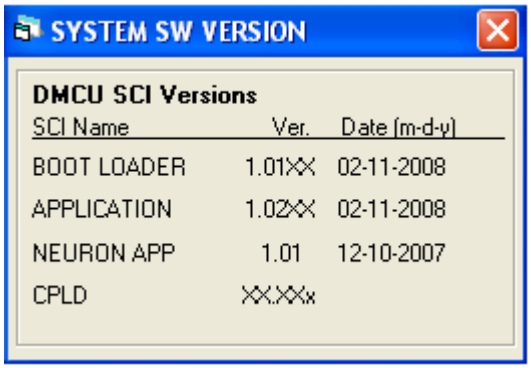


Figure 5-14: System SW Version User Interface

As shown in Figure 5-14, the following information is presented on the System SW Version window:

- DCU Boot Loader Version and Date.
- DCU Application Version and Date.
- Neuron Gateway Application Version and Date.
- CPLD Version.

The System SW Version window also provides a command button:

- Refresh button: To request the DCU to send the current System software versions information and display them on the System SW Version window.

5.3.11.3 Related Information

None

5.3.11.4 Cautions and Warnings

Once proper communication is established between the DCU Application and the PTU Application, there should be no problem observing the System software versions information. Otherwise, double check the status of the RS232 cable and investigate other possible hardware problems.

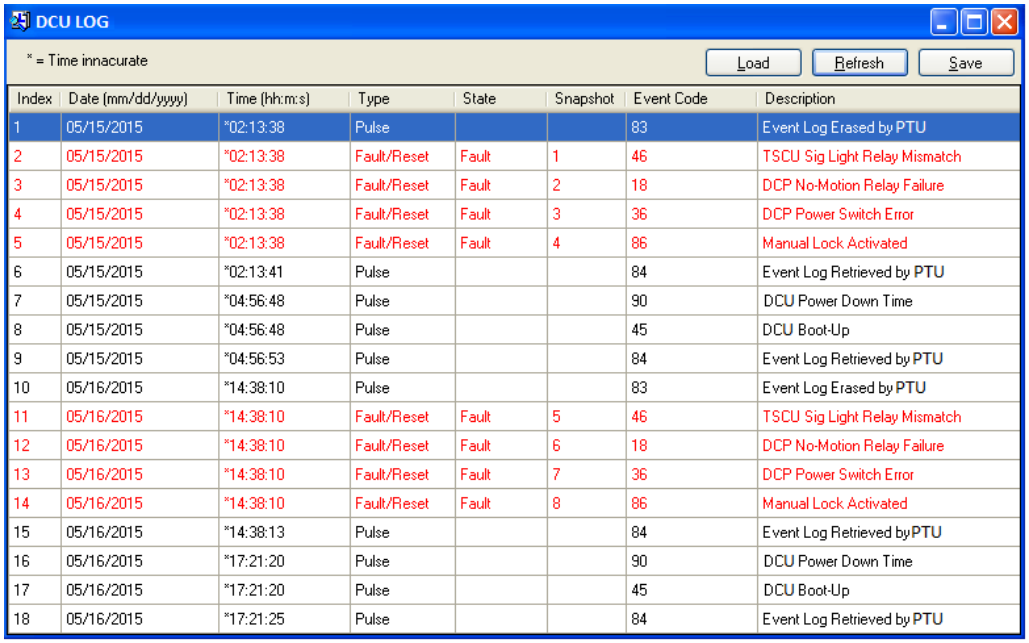
5.3.12 Log Table

5.3.12.1 Scope

The following subsections describe how to display the DCU log information that is currently in the PTU Application database.

5.3.12.2 Method

The DCU Log Table is used to display the DCU logs in a table. To view the event log in the Log Table, click *Maintenance → Log* or click on Log from the PTU Control Menu (Maintenance) or click Log View button on the toolbar.



DCU LOG

* = Time innacurate

Load | Refresh | Save

Index	Date (mm/dd/yyyy)	Time (hh:m:s)	Type	State	Snapshot	Event Code	Description
1	05/15/2015	*02:13:38	Pulse			83	Event Log Erased by PTU
2	05/15/2015	*02:13:38	Fault/Reset	Fault	1	46	TSCU Sig Light Relay Mismatch
3	05/15/2015	*02:13:38	Fault/Reset	Fault	2	18	DCP No-Motion Relay Failure
4	05/15/2015	*02:13:38	Fault/Reset	Fault	3	36	DCP Power Switch Error
5	05/15/2015	*02:13:38	Fault/Reset	Fault	4	86	Manual Lock Activated
6	05/15/2015	*02:13:41	Pulse			84	Event Log Retrieved by PTU
7	05/15/2015	*04:56:48	Pulse			90	DCU Power Down Time
8	05/15/2015	*04:56:48	Pulse			45	DCU Boot-Up
9	05/15/2015	*04:56:53	Pulse			84	Event Log Retrieved by PTU
10	05/16/2015	*14:38:10	Pulse			83	Event Log Erased by PTU
11	05/16/2015	*14:38:10	Fault/Reset	Fault	5	46	TSCU Sig Light Relay Mismatch
12	05/16/2015	*14:38:10	Fault/Reset	Fault	6	18	DCP No-Motion Relay Failure
13	05/16/2015	*14:38:10	Fault/Reset	Fault	7	36	DCP Power Switch Error
14	05/16/2015	*14:38:10	Fault/Reset	Fault	8	86	Manual Lock Activated
15	05/16/2015	*14:38:13	Pulse			84	Event Log Retrieved by PTU
16	05/16/2015	*17:21:20	Pulse			90	DCU Power Down Time
17	05/16/2015	*17:21:20	Pulse			45	DCU Boot-Up
18	05/16/2015	*17:21:25	Pulse			84	Event Log Retrieved by PTU

Figure 5-15: DCU Log Table User Interface[3]

As shown in Figure 5-15, the table has the following columns:

- Index: Sequence number of event
- Date: Date of the event
- Time: Timestamp of the event (local time)
- Type: Event type
- State: Event state
- Snapshot: Related snapshot file number. Logging of certain Event Codes results in the saving of additional diagnostic data prior to and following the event occurrence; this data is stored in an associated snapshot.
- Event Code: Event code number
- Description: Event description

Clicking each column header makes the Log Table re-sort according to the selected header. Right-clicking on an event provides the user options which help reviewing and analyzing information in the table. These options are:

- Mark Event/Remove Mark: Toggle the marking or highlighting of the event.
- Mark Same Event: Highlight the event and all others with the same event code.
- Clear Marks: Remove all markings.

[3] List of events in the snapshot is example for purpose only and may not reflect the Denver Project.

Double-clicking on an event opens the Event Attributes window for the event.

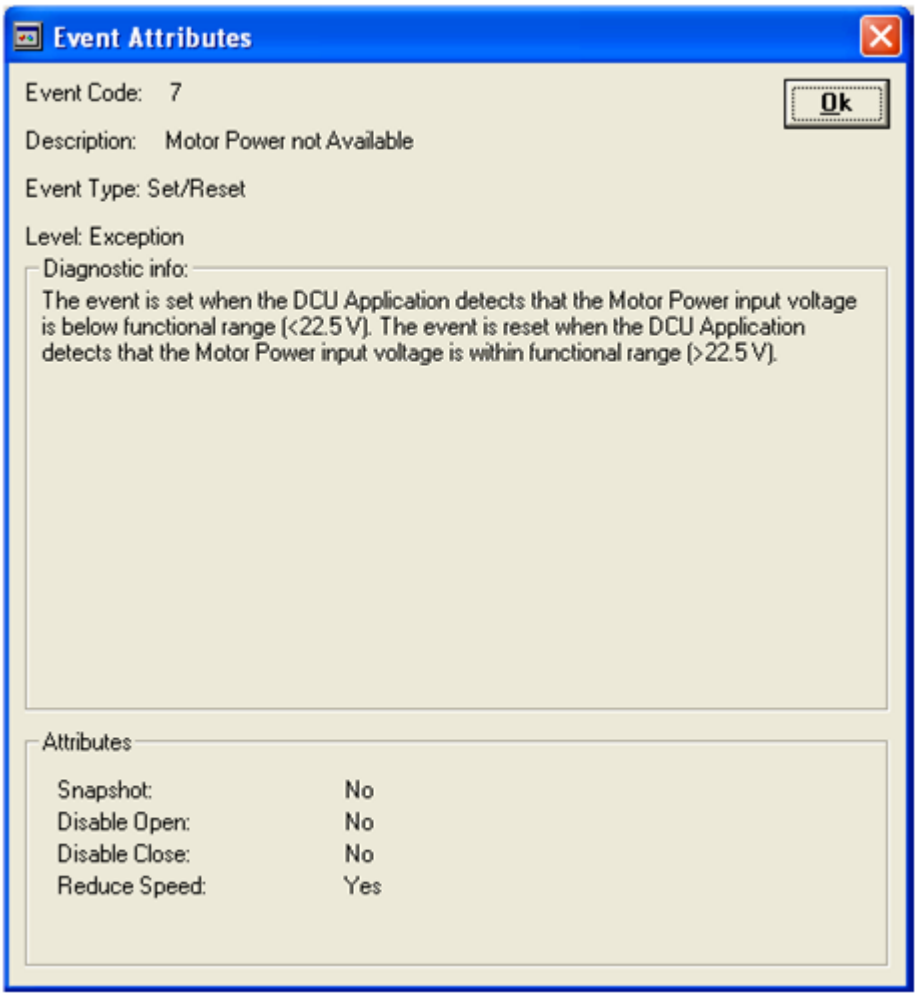


Figure 5-16: Event Attributes window

As shown in Figure 5-16, the Event Attributes window displays the Event Code, Description, Event Type, Event Level, Diagnostic Information, and Attributes.

The DCU Log window also provides command buttons:

- Load button: Open and display the log from a previously saved log text file in the DCU LOG window.
- Refresh button: Perform the Log Download from DCU (see section 5.3.13).
- Save button: Save the current Log to a text file (same as File menu → Save Log As, see section 5.3.1).

5.3.12.3 Related Information

None

5.3.12.4 Cautions and Warnings

The mere launching of the window will only display the log information that is currently in the PTU Application database; the Refresh button in the DCU LOG window, or the Log Download button on the toolbar needs to be clicked to download and display the most current log information. Also, refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.13 Log Download

5.3.13.1 Scope

The following subsections provide information on how to download the latest DCU log information that is used to update the PTU Application's database for display in the Log Table.

5.3.13.2 Method

Click *Maintenance → Other Functions → Log Download* or click the Log Download button from the toolbar.

A message window (Figure 5-17) pops up to show the download process when the PTU Application is performing the Log Download from the DCU. The window is closed after the Log download is finished. The log data is displayed in the Log Table (see Section 5.3.12)

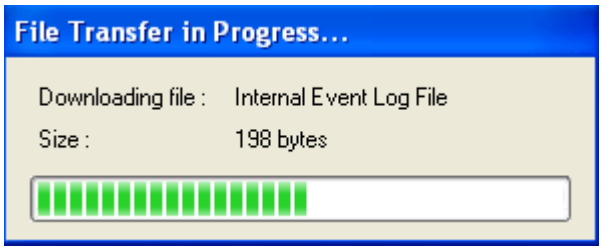


Figure 5-17: Log Download Progress Window

5.3.13.3 Related Information

None

5.3.13.4 Cautions and Warnings

Refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.14 Log Erasure

5.3.14.1 Scope

The following subsections provide information on how to erase the DCU log information stored within the DCU.

5.3.14.2 Method

On the menu, select Administration → Log Erasure or click on “Log Erasure” from the PTU Control Menu (Administration) or click the Log Erasure button on the toolbar. A window (see Figure 5-18) prompts the user to confirm or to cancel the delete action.

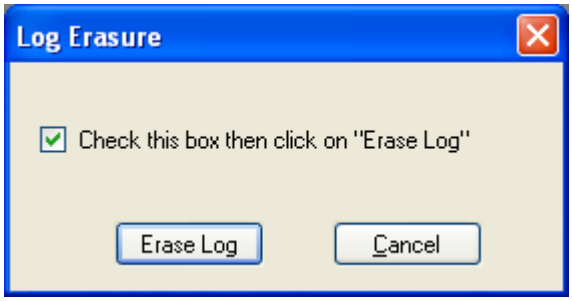


Figure 5-18: Log Erasure Confirmation Window

- Erase Log button: It opens the Delete Event Log File confirmation dialog (see Figure 5-19) box. This button is enabled only after the checkbox is checked.
- Cancel button: Used to cancel the log erasure and close the window.

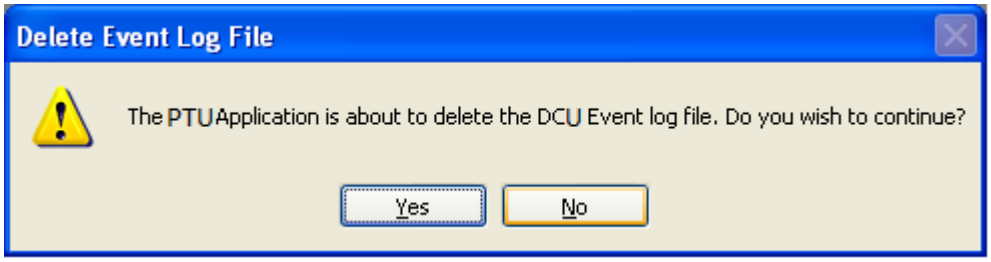


Figure 5-19: Delete Event Log File Confirmation Window

- Yes button: Confirm the Log Deletion
- No button: Cancel the Event Log file deletion.

The Event Log in the DCU is deleted when the user proceeds with the delete action by clicking on the Yes button.

5.3.14.3 Related Information

➢ Administration access password is needed for enabling this function (see Section 5.3.5).

5.3.14.4 Cautions and Warning

Log information erased with this function cannot be recovered; critical diagnostic information could be lost forever. Hence, care should be exercised when using this function.

5.3.15 Snapshot Table

5.3.15.1 Scope

The following subsections describe how to display the DCU Snapshot information that is currently in the PTU Application database. Logging of certain event codes in the event log results in the saving of additional diagnostic data prior to and following the event occurrence; this data is stored in an associated snapshot. The Snapshot Table is used to list all snapshot files of the DCU.

5.3.15.2 Method

To access this, click *Maintenance → Other Functions → Snapshot View* or click on "Other Functions" then "Snapshot View" from the PTU Control Menu (Maintenance) or click the Snapshot View button from the toolbar.

Snapshot Table

No.	Time (m-d-y h:m:s)	Event Code	Rate	# of Parameter	Parameters
1	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
2	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
3	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
4	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
5	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
6	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
7	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
8	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
9	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
10	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
11	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
12	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
13	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
14	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9
15	06:09-2008 09:17:22	38 Motor OC Shutd…	5	57	1,2,3,4,5,6,7,8,9

Figure 5-20: Snapshot Table User Interface[4]

As shown in Figure 5-20, the following snapshot file header information is presented:

- No.: Snapshot number;
- Time: Time Stamp of the snapshot including the date;
- Event Code: Reference Event Code and description;
- Rate: Sampling Rate;
- # of Parameter: The number of parameters in the file;
- Parameters: List of parameters.

To look at the snapshot contents, double-click on the snapshot file. The PTU Application opens the Snapshot View window to show the details of that snapshot (see section 5.3.17).

5.3.15.3 Related Information

None

5.3.15.4 Cautions and Warnings

The mere launching of the window will only display the Snapshot information that is currently in the PTU Application database; the Snapshot information needs to be downloaded to display the most current Snapshot information (see Section 5.3.16).

[4] List of events in the snapshot is for example purpose only and may not reflect the Denver Project.

5.3.16 Snapshot Download

5.3.16.1 Scope

The following subsections provide information on how to download the latest DCU Snapshot Files.

5.3.16.2 Method

Click *Maintenance* → *Other Functions* → *Snapshot Download* or click the Snapshot Download button from the toolbar.

A message window (Figure 5-21) pops up for each snapshot to show the download process when the PTU Application is performing the Snapshot Download from the DCU. The window is closed after the Snapshot Download is finished. The snapshot data are displayed on the Snapshot Table (see Section 5.3.15).

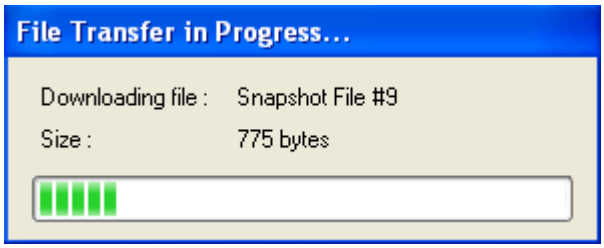


Figure 5-21: Snapshot Download Progress Window

5.3.16.3 Related Information

None

5.3.16.4 Cautions and Warnings

Refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.17 Snapshot View

5.3.17.1 Scope

The following subsections provide information on how to display the detailed information contained within a Snapshot file entry.

5.3.17.2 Method

Double-click the snapshot item on the Snapshot Table (Section 5.3.15), the Snapshot View window will open and the snapshot item is displayed on the Snapshot View window.

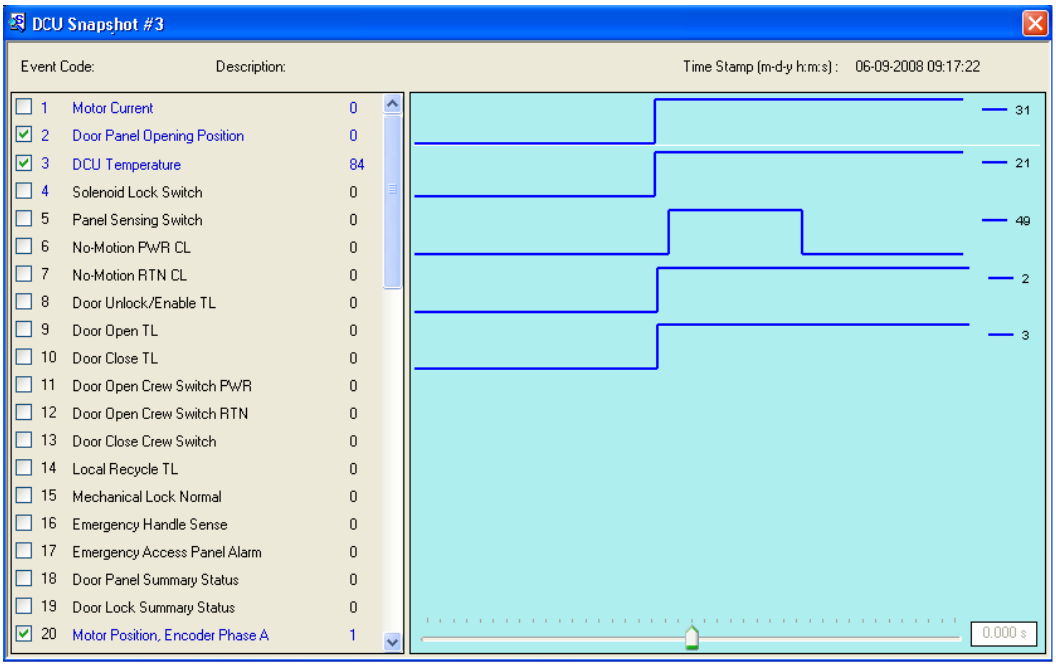


Figure 5-22: Snapshot View User Interface[5]

The Snapshot View allows the user to explore the snapshot contents. As shown in Figure 5-22, the following information is presented:

- Snapshot file header: Reference event code, description, and time stamp;
- Parameters list with checkboxes: The parameter ID and description list that are recorded in the snapshot file are displayed on the left side of the window. A parameter description in blue text indicates that the value of the parameter is not the same for all the duration of the snapshot, meaning that the signal is probably more interesting. The parameters of interest can be selected by placing checkmarks in the checkboxes;
- Parameter value display window: Display the selected parameter values by using graphics on the right side of the window. The graphic starts from the first valid record of the snapshot;
- Scrollbar: The scrollbar at the bottom is used to scroll through the time scale of the graphic. The value shown in the Parameters list is the value at the scrollbar position;
- Time box: The scrollbar position is displayed in a text box at the bottom right of the window.

If there are more parameters than the window can fit, a scroll bar will be presented at the right side of the Parameters list. Move the scroll bar to browse the parameters list. Placing or removing a check in the parameter check box will add or remove the graphic from the graphic window. A scroll

[5] List of signals in the snapshot is for example purpose only and may not reflect the Denver Project's signals.

bar is also presented if required for the graphic side of the window if many parameters are selected. By moving the snapshot time scroll bar, the user is able to browse all snapshot values.

5.3.17.3 Related Information

None

5.3.17.4 Cautions and Warnings

None

5.3.18 Snapshot Erasure

5.3.18.1 Scope

The following subsections provide information on how to erase the Snapshot file maintained within the DCU.

5.3.18.2 Method

On the menu, click *Administration → Other Functions → Snapshot Erasure* or click on "Other Functions" then "Snapshot Erasure" from the PTU Control Menu (Administration) or click on the Snapshot Erasure button on the toolbar. A window (see Figure 5-23) prompts the user to select the snapshots needed to be deleted. After clicking on the Delete button in the Snapshot Erasure window, a confirmation window (see Figure 5-24) prompts the user to confirm or to cancel the delete action.

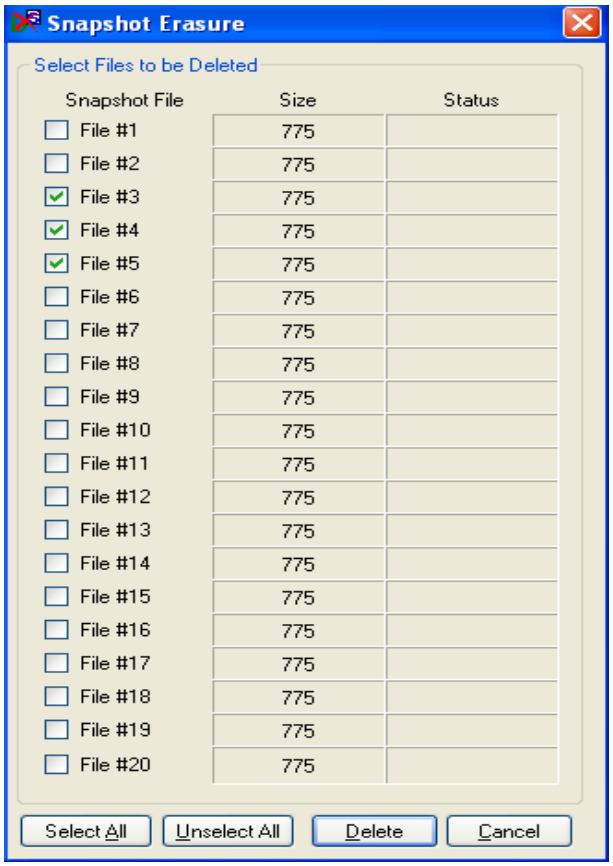


Figure 5-23: Snapshot Erasure Selection User Interface

The following information is presented in the window:

- DCU Snapshot File #n checkboxes: Select the DCU Snapshot File to delete;
- Size: The size of the related snapshot file;
- Status: When the user proceeds with the Snapshot File delete action, the Status changes to "Deleted";
- Select All button: Select all snapshots;
- Unselect All button: Unselect all snapshot;
- Delete button: Proceed with the DCU Snapshot delete action;
- Cancel button: Cancel the DCU Snapshot delete action.

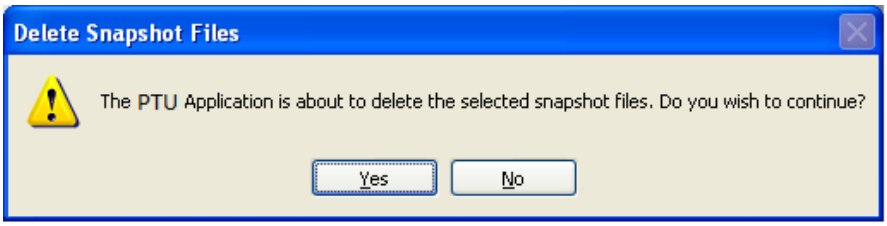


Figure 5-24: Snapshot Erasure Confirmation Message Window

The following information is presented in the window

- Yes button: Confirm the DCU Snapshot delete action;
- No button: Cancel the DCU Snapshot delete action.

The deleted Snapshots will be shown on the selection window with a Status of "Deleted".

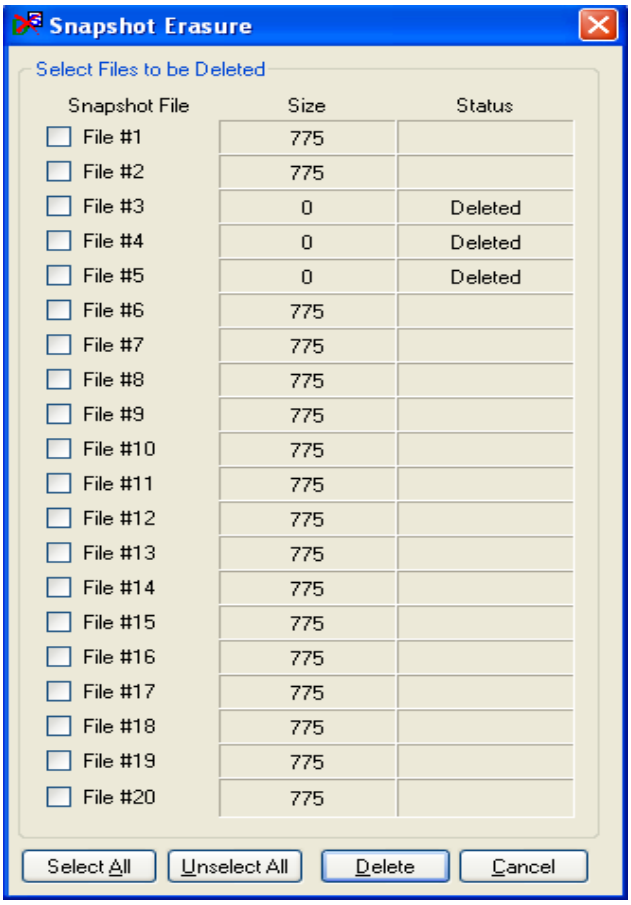


Figure 5-25: Snapshot Erasure Status

5.3.18.3 Related Information

➢ Administration access password is needed for this function (see Section 5.3.5).

5.3.18.4 Cautions and Warning

Snapshot information erased with this function cannot be recovered; critical diagnostic information could be lost forever. Hence, care should be exercised when using this function.

5.3.19 Graph View

5.3.19.1 Scope

The following subsections describe how to display the door travel graph information that is currently in the PTU Application database.

5.3.19.2 Method

To open the Door Move Graph window, click *Maintenance → Other Functions → Graph View* or click on "Other Functions" then "Graph View" from the PTU Control Menu (Maintenance) or click the Graph View button on the toolbar.

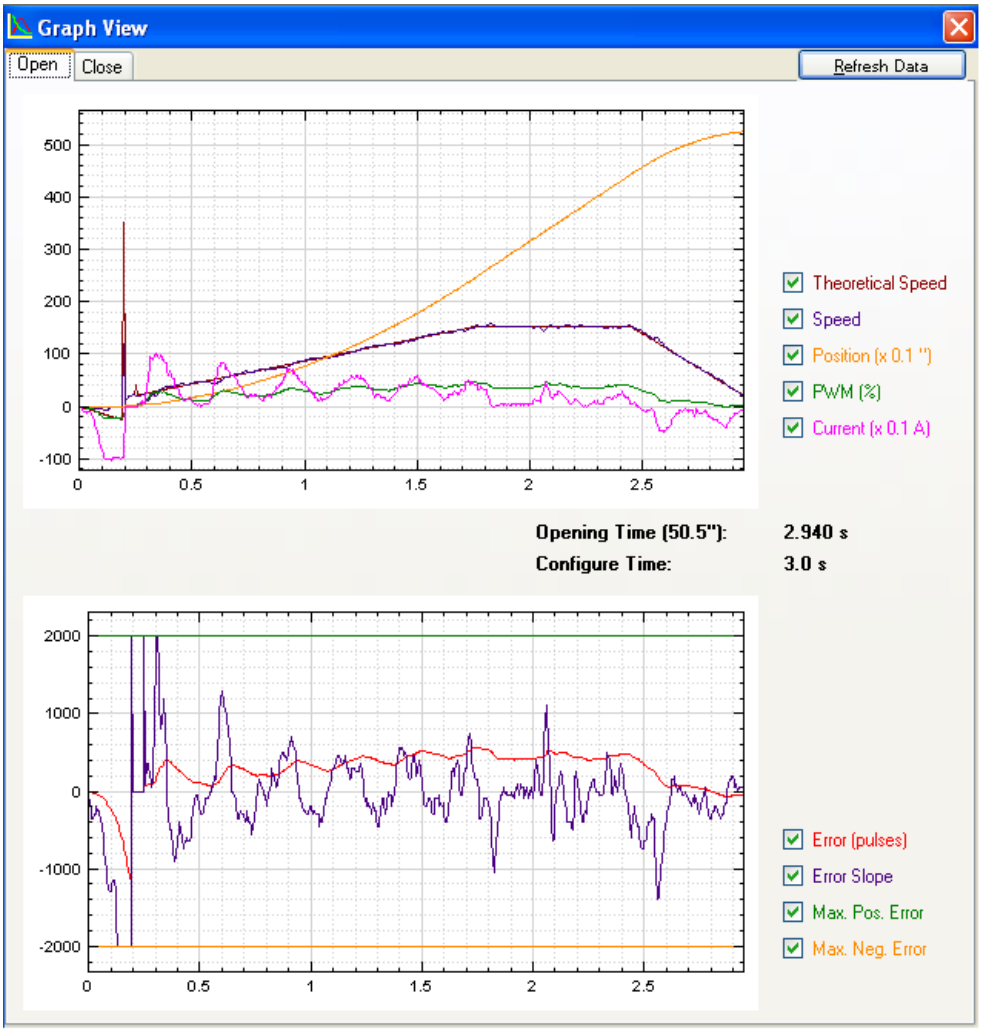


Figure 5-26: Door Move Graph User Interface

The Graph View window (Figure 5-26) shows door movement information.

The top graph displays:

- Theoretical speed (intended)
- Measured speed
- Position
- PWM (%)
- Current

The bottom graph displays:

- Error (pulses)
- Error (slope)
- Max Pos. Error
- Max Neg. Error
- Opening/Closing time
- Configured Opening/Closing time

The Tabs Open and Close are used to switch between the corresponding graphs.

The Door Move Graph window also provides a command button:

- Refresh Data button: Perform the Door Move Graph Download from the DCU (see Section 5.3.20).

5.3.19.3 Related Information

None

5.3.19.4 Cautions and Warnings

The mere launching of the window will only display the Graph information that is currently in the PTU Application database; the Graph information needs to be downloaded to display the most current information (see Section 5.3.20).

5.3.20 Graph Download

5.3.20.1 Scope

The following subsections provide information on how to download the latest door graph information that is used to update the PTU Application's database for display in the Graph View window.

5.3.20.2 Method

To initiate a Door Graph download, click *Maintenance → Other Functions → Graph Download* or click on "Other Functions" then "Graph Download" from the PTU Control Menu (Maintenance) or click the Graph Download button on toolbar.

The Graph View window (Section 5.3.19) is opened and a message window (Figure 5-27) pops up to show the download process when the PTU Application is performing the Door Move Graph Download from the DCU. The progress window is closed after the Door Move Graph Download is finished. The downloaded graph information is then displayed in the Graph View window.

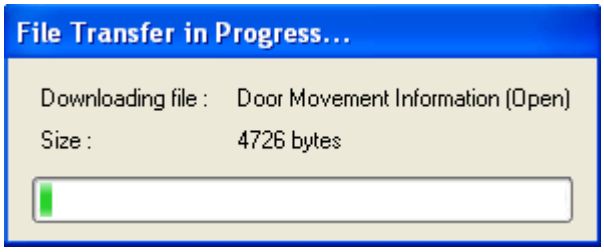


Figure 5-27: Door Move Graph Download Progress Window

5.3.20.3 Related Information

- Should there have been no door movement since the last DCU reboot, there would be no meaningful graph information to display.

5.3.20.4 Cautions and Warnings

Refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

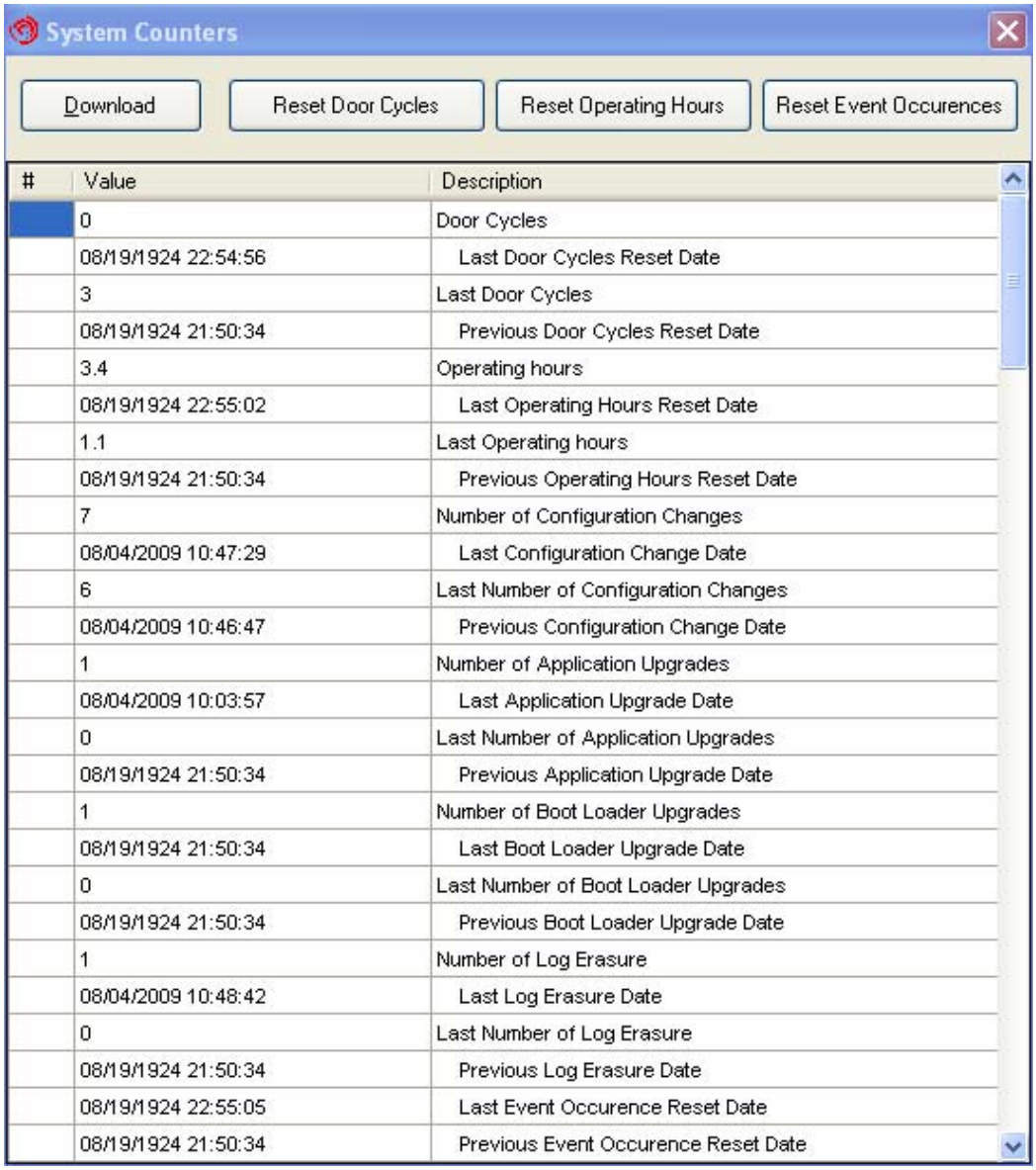
5.3.21 System Counters

5.3.21.1 Scope

The following subsections provide information on how to display and reset the counters maintained by the DCU for the number of door cycles, the number of operating hours and the number of times DCU Log events occurred.

5.3.21.2 Method

On the menu, click *Maintenance → Other Functions → System Counters View* or click on “Other Functions” then “System Counters View” from the PTU Control Menu (Maintenance) . A window as shown in Figure 5-28 appears.



#	Value	Description
	0	Door Cycles
	08/19/1924 22:54:56	Last Door Cycles Reset Date
	3	Last Door Cycles
	08/19/1924 21:50:34	Previous Door Cycles Reset Date
	3.4	Operating hours
	08/19/1924 22:55:02	Last Operating Hours Reset Date
	1.1	Last Operating hours
	08/19/1924 21:50:34	Previous Operating Hours Reset Date
	7	Number of Configuration Changes
	08/04/2009 10:47:29	Last Configuration Change Date
	6	Last Number of Configuration Changes
	08/04/2009 10:46:47	Previous Configuration Change Date
	1	Number of Application Upgrades
	08/04/2009 10:03:57	Last Application Upgrade Date
	0	Last Number of Application Upgrades
	08/19/1924 21:50:34	Previous Application Upgrade Date
	1	Number of Boot Loader Upgrades
	08/19/1924 21:50:34	Last Boot Loader Upgrade Date
	0	Last Number of Boot Loader Upgrades
	08/19/1924 21:50:34	Previous Boot Loader Upgrade Date
	1	Number of Log Erasure
	08/04/2009 10:48:42	Last Log Erasure Date
	0	Last Number of Log Erasure
	08/19/1924 21:50:34	Previous Log Erasure Date
	08/19/1924 22:55:05	Last Event Occurence Reset Date
	08/19/1924 21:50:34	Previous Event Occurence Reset Date

Figure 5-28: System Counters

The "Value" column in System Counters view contains counts, date and time or number of hours. The "Description" column describes what the "Value" column represents. The "#" column is used

for event occurrences and represents the Event Code. The dates for a counter reset, a configuration change, an application upgrade, and a Log Erasure are also displayed. The following are included in the System Counters:

- Counters for the number of times a given event occurred since the last time the counters were reset
- Counter for the number of times the door has cycled (opened and closed) since the last time the counter was reset
- Counter for the number of hours the DCU has been operational (powered On) since the last time the counter was reset

Upon the initial launching of the window when System Counters View feature is used, the values displayed are those in the PTU database. If current values of the counters in the DCU are to be displayed, click the Download button. Alternately, if it is desired that the values displayed upon the initial launching of the window be the current values in the DCU, use the main menu option Maintenance → Other Functions → System Counters Download or click on “Other Functions” then “System Counters Download” from the PTU Control Menu (Maintenance) .

The counters can be reset by clicking their corresponding reset buttons. For example, if the counters for the number of times the respective events occurred are to be reset, click the Reset Event Occurrences button. The reset buttons are enabled only if the administration access password has been entered.

5.3.21.3 Related Information

- Administration access password is needed for enabling the resetting of System Counters (see Section 5.3.5).

5.3.21.4 Cautions and Warning

Information reset or erased with this function cannot be recovered; critical diagnostic information could be lost forever. Hence, care should be exercised when using this function.

5.3.22 DCU Configure

5.3.22.1 Scope

The following subsections provide information on how to view and change DCU parameters that affect door behavior.

5.3.22.2 Method

To configure the DCU, click *Administration → Other Functions → DCU Configure* or click on "Other Functions" then "DCU Configure" from the PTU Control Menu (Administration). Administration functions are available only if the administration access password is entered after PTU Application start-up.

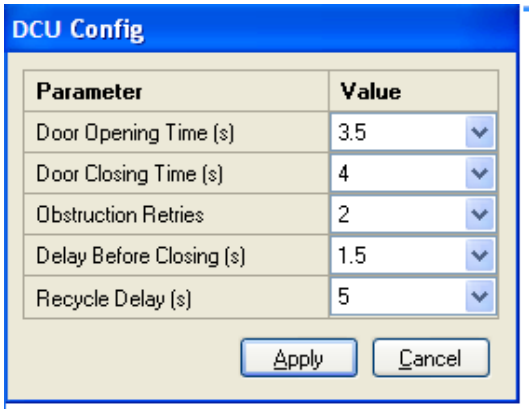


Figure 5-29: DCU Configure User Interface[6]

The following can be configured with the PTU Application:

Table 5-1: DCU Configuration

Description	Valid Values	Default	Units
Opening Time	2.0, 2.5, 3.0 seconds	**2.0**	**Second(s)**
Closing Time	2.0, 2.5, 3.0 seconds	2.5	Second(s)
Obstruction Retries (Recycles)	0 to 5 attempts	3	Attempt(s)
Obstruction re-open distance	0, 3, 5, 10, 15, 20, and 25 inches	3	Inch(es)
Delay before reclosing	0 to 5 in 1 second increments	2	Second(s)
Delay before door closing	0 to 5 in 1 second increments	3	Second(s)
Door open duration for Passenger Door Enable	0 to 10 in 1 minute increments	5	Minute(s)

5.3.22.3 Related Information

- Administration access password is needed for enabling configuration changes (see Section 5.3.5).

5.3.22.4 Cautions and Warnings

Adjustment of the parameters could also affect operational delays.

[6] The information shown in this figure is preliminary.

5.3.23 DCU Time

5.3.23.1 Scope

The following subsections provide information on how to retrieve and change the current time maintained by the DCU.

5.3.23.2 Method

To retrieve the DCU time, click *Tools → DCU Time* from the PTU menu. The DCU Time window is used to retrieve and update the current time in the DCU.

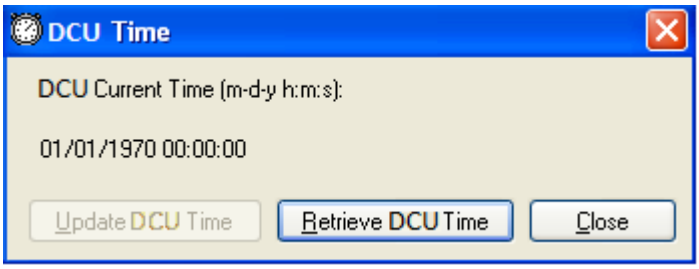


Figure 5-30: DCU Time User Interface

As shown in Figure 5-30, there are three function buttons on this window:

- Update DCU Time button: Send the computer time to the DCU to update its internal time value. This button is enabled only if the administration access password has been entered and RS232 communication device is being used.
- Retrieve DCU Time button: Refresh the DCU Time display by getting the time value from the DCU;
- Close button: Close the DCU Time window.

5.3.23.3 Related Information

➢ Administration access password is needed for enabling DCU Time updates (see Section 5.3.5).

5.3.23.4 Cautions and Warnings

Make sure the date and time on the PTU computer is correct before doing a DCU Time update.

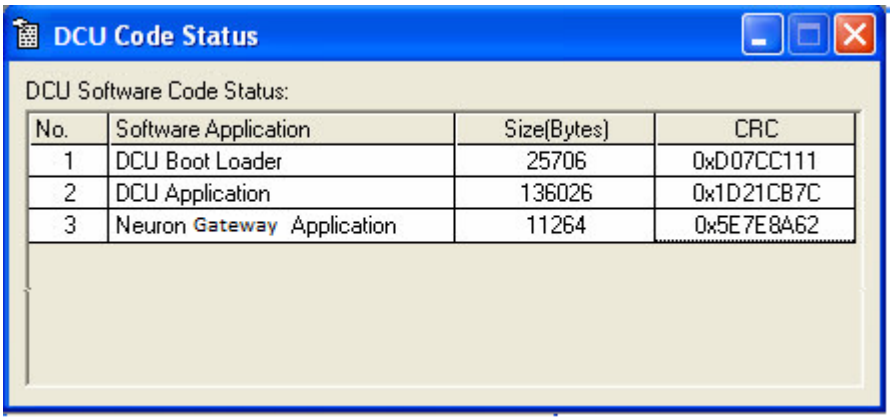
5.3.24 DCU Code Status

5.3.24.1 Scope

The following subsections provide information on how to view the size and CRC32 for the DCU Application and DCU Boot Loader.

5.3.24.2 Method

To retrieve the DCU Code Status, click *Tools → DCU Code Status* from the PTU menu.



No.	Software Application	Size(Bytes)	CRC
1	DCU Boot Loader	25706	0xD07CC111
2	DCU Application	136026	0x1D21CB7C
3	Neuron Gateway Application	11264	0x5E7E8A62

Figure 5-31: DCU Code Status Window

As shown in Figure 5-31, the PTU DCU Code Status window is used to retrieve the following information:

- DCU Boot Loader size and CRC
- DCU Application size and CRC
- Neuron Gateway Application size and CRC

5.3.24.3 Related Information

None

5.3.24.4 Cautions and Warning

None

5.3.25 SW Download DCU Application

5.3.25.1 Scope

The following subsections provide information on how to download or update the DCU Application software.

5.3.25.2 Method

To download the DCU Application into the DCU, the PTU user has to select the DCU Application (Binary file) first. Then the PTU is able to use that DCU Application to download it into the DCU.

Click *Administration → SW Download* or click on "SW Download" from the PTU Control Menu (Administration). Select the DCU Application binary file in the list.

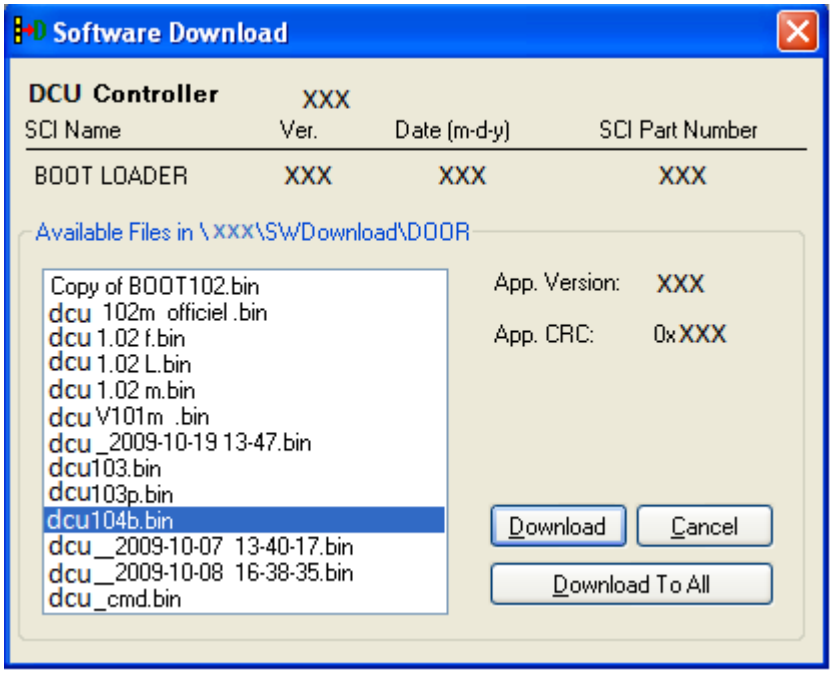


Figure 5-32: Software Download Window

After the DCU Application is selected, the DCU Application CRC is calculated and the version number displayed to help the user verify if the DCU Application is corrupted.

Click the Download button (Figure 5-32) to start DCU upgrade of the connected DCU.

The user confirmation window (Figure 5-33) pops up after the user requests to download the DCU Application with the selected file into the DCU.

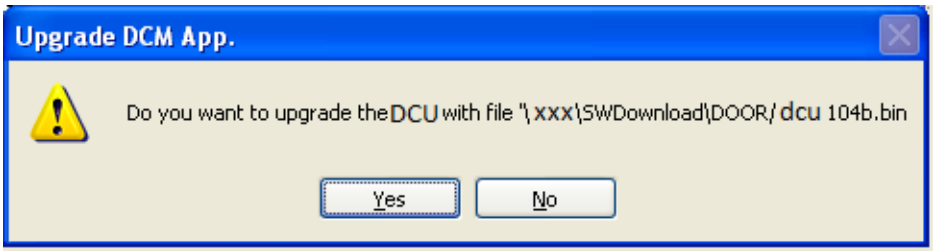


Figure 5-33: DCU Application Upgrade Confirmation Message Window

During the file transfer, the progress state is monitored (Figure 5-34).

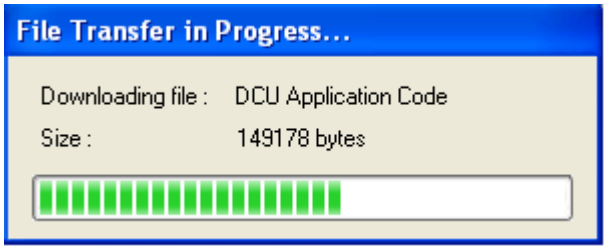


Figure 5-34: DCU Application Download Progress Window

If no error occurs during the transfer, the progress window is closed and the PTU is logged out after the upgrade is completed.

5.3.25.3 Related Information

- Administration access password is needed for enabling the Download buttons in the Software Download window (see Section 5.3.5).

5.3.25.4 Cautions and Warnings

It is important that only the officially released version of the software file be placed in the software download directory to ensure proper configuration control of the DCU. Otherwise, improper version could be accidentally uploaded.

5.3.26 SW Download DCU Boot Loader

5.3.26.1 Scope

The following subsections provide information on how to download or update the DCU Boot Loader software.

5.3.26.2 Method

To download the Boot Loader Application into the DCU, execute the following steps:

1. Turn OFF the DCU.

2. Make sure the DCU is in Bootstrap Mode.

3. Select *Administration → Other Functions → Load Flash → Send Flash File* or click on "Other Functions" then "Send Flash File" from the PTU Control Menu (Administration) to start the download of the DCU Boot Loader Application. The Flash File Transfer window as shown in Figure 5-35 is opened showing the upgrade status with messages.

4. Turn ON DCU.

5. Once the DCU Boot Loader upgrade process is finished, verify that the uploading was successful by checking for the messages "Flash file transferred", "End Upgrade!!!", and "Operation has been completed." in the Flash File Transfer window. If the uploading was successful, remove the Bootstrap Mode on the DCU and reboot.

6. If the uploading failed, redo steps 3 and 5.

Figure 5-35: DCU Boot Loader Download Status Window

5.3.26.3 Related Information

- Administration access password is needed for enabling this function (see Section 5.3.5).

5.3.26.4 Cautions and Warning

It is extremely important that only the officially released version of the software file be placed in the software download directory (\DENVERPTU\SWDownload\DOOR) to ensure proper configuration control of the DCU. Otherwise, door operation and safety could be compromised.

After the upgrade, the DCU Boot Loader software version, date, size and CRC can be verified by using the System SW Version (see Section 5.3.11) and DCU Code Status (see Section 5.3.24) features.

5.3.27 SW Download DCU Neuron Application

5.3.27.1 Scope

The following sub-sections provide information on how to download or update the DCU Neuron Application software in the DCU. This function uses a command prompt based program from Echelon Corporation called NodeUtil. When the function is initiated as described below, a standalone window shown in Figure 5-36 appears when NodeUtil is launched. Additional help related to NodeUtil, beyond what is described herein, is available by pressing the 'h' key on the keyboard.

5.3.27.2 Method

To download the DCU Neuron Application into the DCU, execute the following steps:

1. Select Administration → Other Functions → NodeUtil (for Neuron Upgrade) or click on "Other Functions" then "NodeUtil" from the PTU Control Menu (Administration) to open NodeUtil software tool (see Figure 5-36);
2. Send the Service Pin from the DCU by momentarily either (a) moving the switch mounted on the special PTU cable (Vapor part # 200620378) to the Boot Strap mode position, or (b) opening up the DCU cover and shorting pins 39 and 40 on test connector 2 (TC-2). NodeUtil will receive a message from the DCU (see Figure 5-37).
3. On the Command Prompt window, type "G" to go to device menu. Type 1 and press [Enter] to chose node 1 (node 1 is the DCU that just sent the service pin). The menu of the Figure 5-38 will appear.
4. Type "Y" to Download Neuron Executable. Type the full path and the name of the file to program 2005L0095-00R-*x*.nxe (where the *x* stands for the release letter of the item) extracted from the DCU Neuron Application package (2005L0095-00R-*x*.zip) and press [Enter]. Note: the file is located in *z*:\DENVER\SWDownload\DOORV\, where the *z* stands for the drive letter (see Figure 5-39).
5. When the file is successfully downloaded, press "Y" to continue (see Figure 5-40).
6. Press "E" twice to exit NodeUtil.

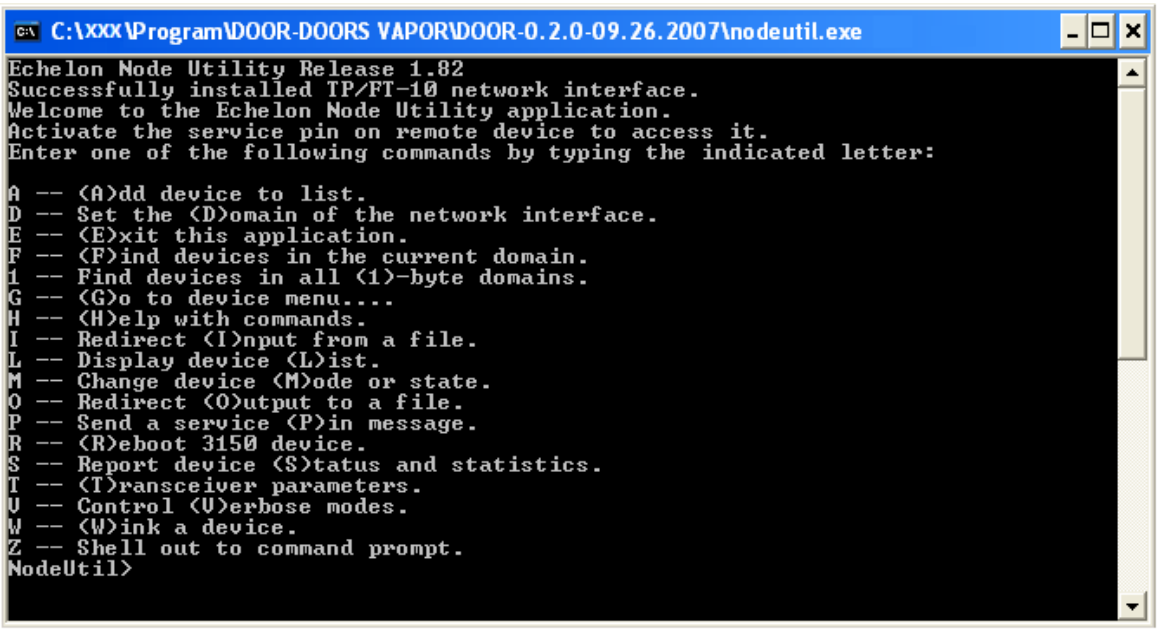


Figure 5-36: NodeUtil Main Menu User Interface

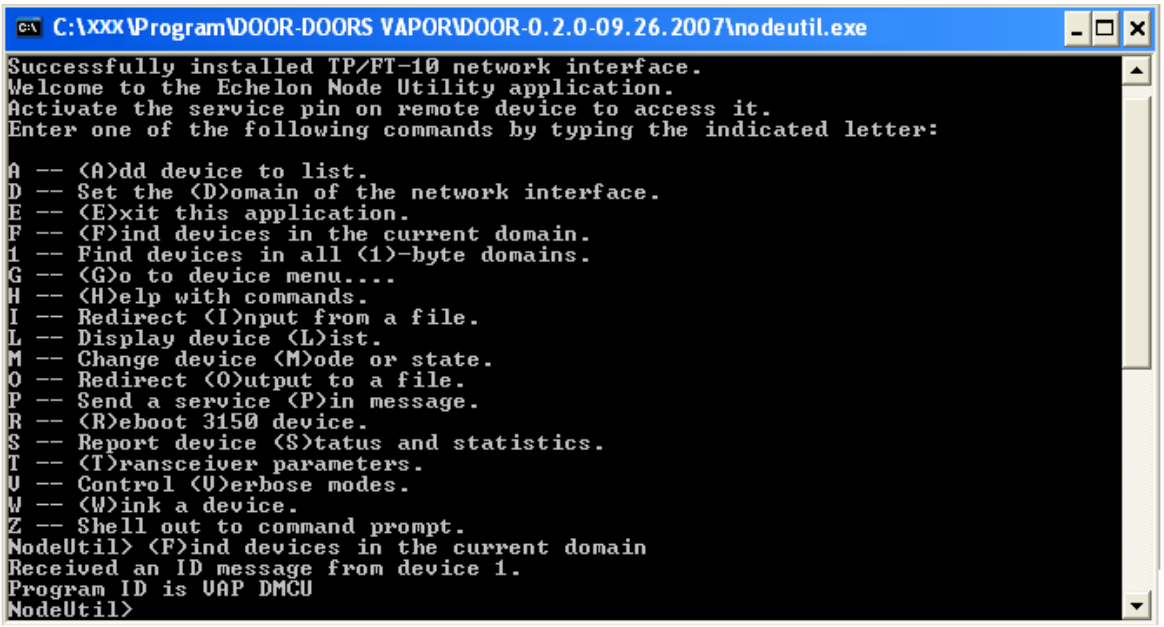


Figure 5-37: NodeUtil Service Pin Message Window

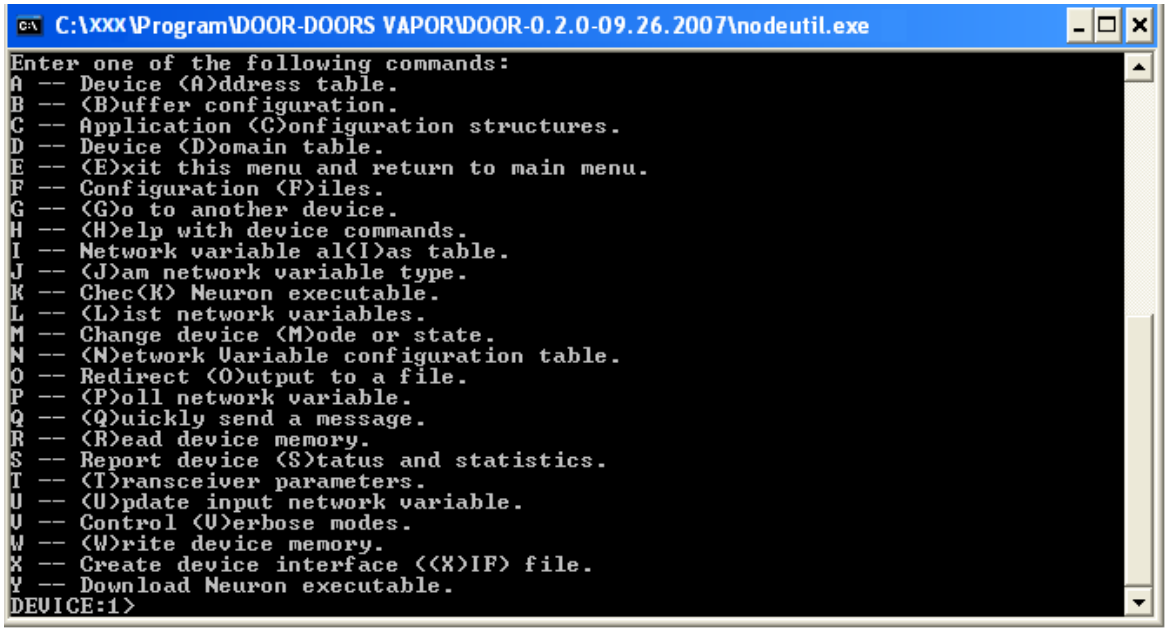


Figure 5-38: NodeUtil Device Menu User Interface

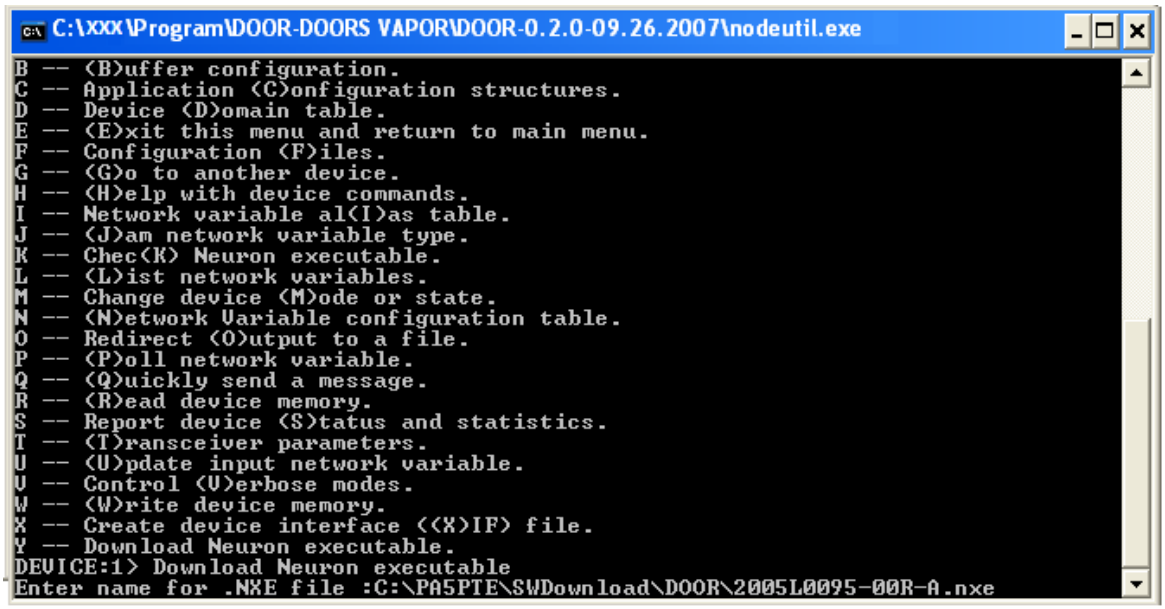


Figure 5-39: NodeUtil Download Executable User Interface

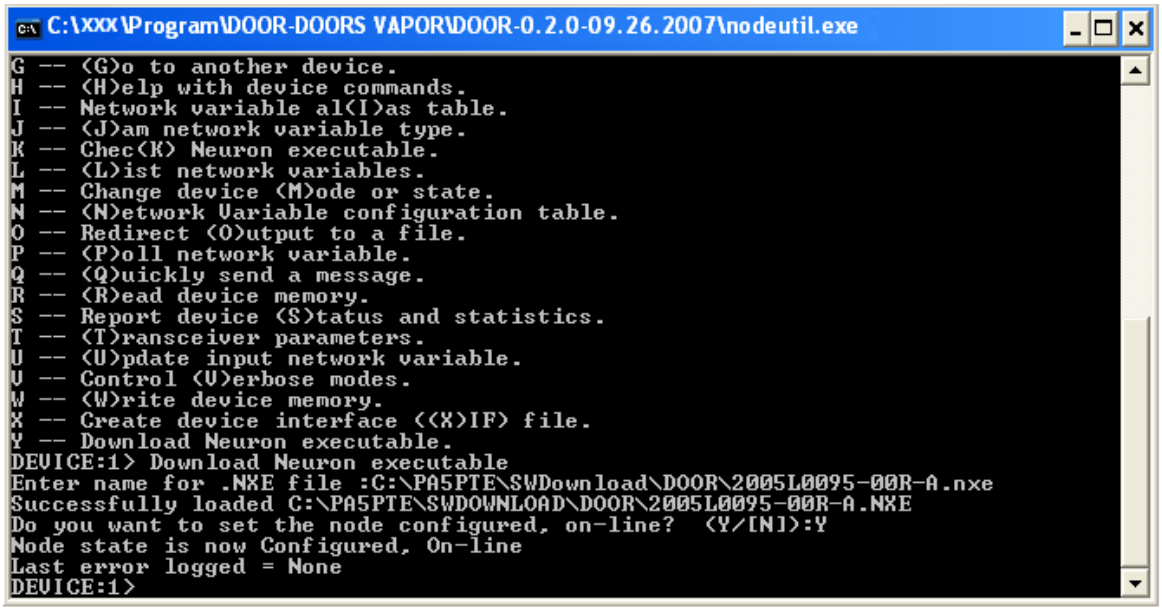


Figure 5-40: NodeUtil Download Successful Message Window

5.3.27.3 Related Information

- Administration access password is needed for enabling the Download buttons in the Software Download window (see Section 5.3.5).

5.3.27.4 Cautions and Warnings

It is extremely important that only the officially released version of the software file be placed in the software download directory to ensure proper configuration control of the DCU. Otherwise, door operation and safety could be compromised.

5.3.28 Operating Parameter List

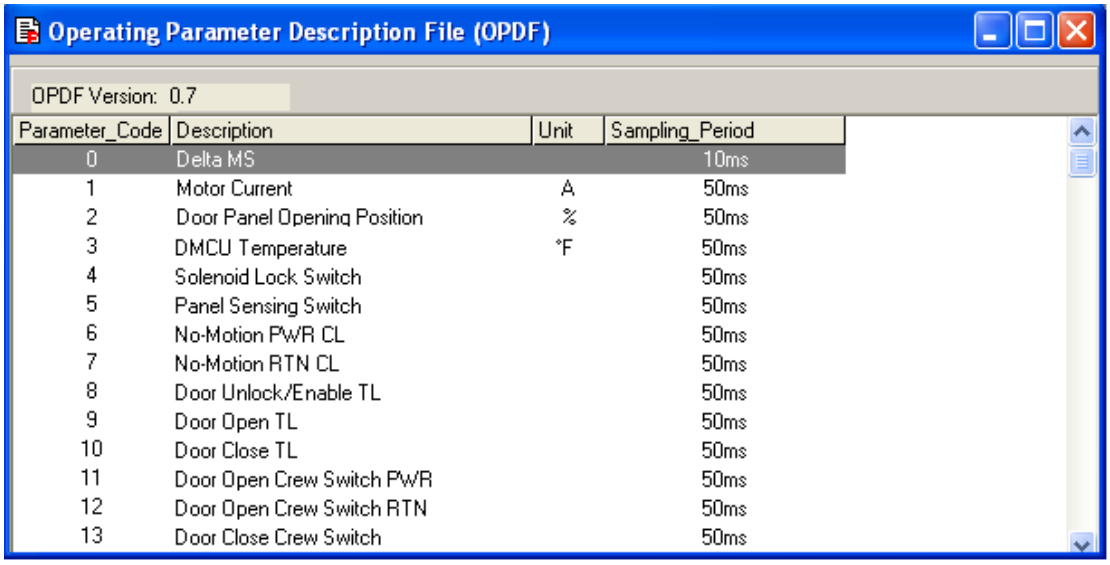
5.3.28.1 Scope

The following subsections provide information on how to view the Operating Parameter Description File information that is in the PTU Application's database.

5.3.28.2 Method

To open the Operating Parameter Description File window, click *Tools → Operating Parameters List* from the PTU menu.

The Operating Parameter Description File (OPDF) window displays the descriptions of all the operating parameters used in the DCU. The OPDF version is shown at the top of the window.



Parameter_Code	Description	Unit	Sampling_Period
0	Delta MS		10ms
1	Motor Current	A	50ms
2	Door Panel Opening Position	%	50ms
3	DMCU Temperature	°F	50ms
4	Solenoid Lock Switch		50ms
5	Panel Sensing Switch		50ms
6	No-Motion PWR CL		50ms
7	No-Motion RTN CL		50ms
8	Door Unlock/Enable TL		50ms
9	Door Open TL		50ms
10	Door Close TL		50ms
11	Door Open Crew Switch PWR		50ms
12	Door Open Crew Switch RTN		50ms
13	Door Close Crew Switch		50ms

Figure 5-41: OPDF User Interface[7]

As shown in Figure 5-41, the following file headers information is presented:

- Parameter Code: Sequential number for the Parameter
- Description: Description of the parameter;
- Unit: Unit of measure for parameter data.
- Sampling Period: Frequency at which sampling occurs.

5.3.28.3 Related Information

None

5.3.28.4 Cautions and Warnings

The mere launching of the window will only display the OPDF that is currently in the PTU Application database; the OPDF needs to be downloaded from the DCU to display the most current file in the DCU (see Section 5.3.30).

[7] List of signals in the snapshot is for example purpose only and may not reflect the Denver Project's Signals

5.3.29 Event Description List

5.3.29.1 Scope

The following subsections provide information on how to view the Event Description File information that is in the PTU Application's database.

5.3.29.2 Method

To open the Event Description File window, click *Tools → Event Descriptions List*.

The Event Description File (EDF) window displays the descriptions of all the events monitored and logged by the DCU. The EDF version is shown at the top of the window.

Event Description File (EDF)

EDF Version: 0.5

Event Code	Attributes	Description	Type	Diagnostic
1	0x02A6	Door Open Trainline Mismatch	Fault/Reset	Door Open trainline is activated without activation o...
2	0x02A5	Solenoid Lock Supply Failure	Fault/Reset	The Solenoid Lock Supply is available when it shou...
3	0x02A5	Motor Open Supply Failure	Fault/Reset	The Motor Open Supply is available when it should ...
4	0x02A6	No-Motion Trainline Mismatch	Fault/Reset	Discrepancy between No-Motion #1 and #2 trainlin...
5	0x0026	Door Override TL Mismatch	Fault/Reset	Discrepancy between Door Override #1 and #2 trai...
6	0x0086	Int. CS Enable w/o Open Signal	Fault/Reset	Interior Crew Switch Door Open Enable is activated...
7	0x0206	Ext. CS Enable w/o Open Signal	Fault/Reset	Exterior Crew Switch Door Open Enable is activate...
8	0x0286	CS Half Open w/o Enable Signal	Fault/Reset	Crew Switch Door Half Open is activated without a...
9	0x0286	CS Full Open w/o Enable Signal	Fault/Reset	Crew Switch Door Full Open is activated without ac...
10	0x0085	Validated Open ICS Error	Fault/Reset	Discrepancy between CPLD and CPU in monitoring...
11	0x0205	Validated Open ECS Error	Fault/Reset	Discrepancy between CPLD and CPU in monitoring...
12	0x0006	Panel Sensing SW Mismatch	Fault/Reset	Discrepancy between Panel Sensing Switches #1 ...
13	0x0006	Solenoid Lock SW Mismatch	Fault/Reset	Discrepancy between Solenoid Lock Switches #1 ...
14	0x0806	High Door Operator Temperature	Fault/Reset	Door Operator Temperature is above 185øF (85 øC).
15	0x0004	Low Door Op Temp (not used)	Fault/Reset	- Not Required Anymore -
16	0x0006	TSCU Signal Light Relay Fail	Fault/Reset	TSCU Signal Light Relays #1 and #2 indicate that t...
17	0x0006	DCP Door Override Relay Fail	Fault/Reset	DCP Door Override Relay #1 or #2 does not follow ...
18	0x0006	DCP No-Motion Relay Failure	Fault/Reset	DCP No-Motion Relay #1 or #2 does not follow No-...
19	0x0005	Motion Controller Module Error	Fault/Reset	Communication error between software modules "M...
20	0x07E5	NVRAM Memory Error	Fault/Reset	NVRAM is defective or access is incorrect.
21	0x07E5	SRAM Memory Error	Fault/Reset	SRAM is defective or access is incorrect.
22	0x07E5	DCU Application CRC32 Error	Fault/Reset	DCU Application Program is corrupted or access to ...
23	0x0005	DCU BootLoader App CRC32 Error	Fault/Reset	DCU Boot Loader Application Program is corrupted ...
24	0x0006	Door Locking Time too long	Fault/Reset	The door is unable to lock within 1 second from the...
25	0x0006	Door Unlocking Timeout	Fault/Reset	The door is unable to unlock within 1.4 seconds.

Figure 5-42: EDF User Interface[8]

As shown in Figure 5-42, the following file header information is presented:

- Event Code;
- Attributes;
- Description: Description of the event;
- Type: Event type (Fault/Reset or Pulse);
- Diagnostic: Information to help diagnose problems.

[8] List of Events in the snapshot is for example purpose only and may not reflect the Denver Project.

Double-clicking on an event provides the user with more information about it (Figure 5-43).

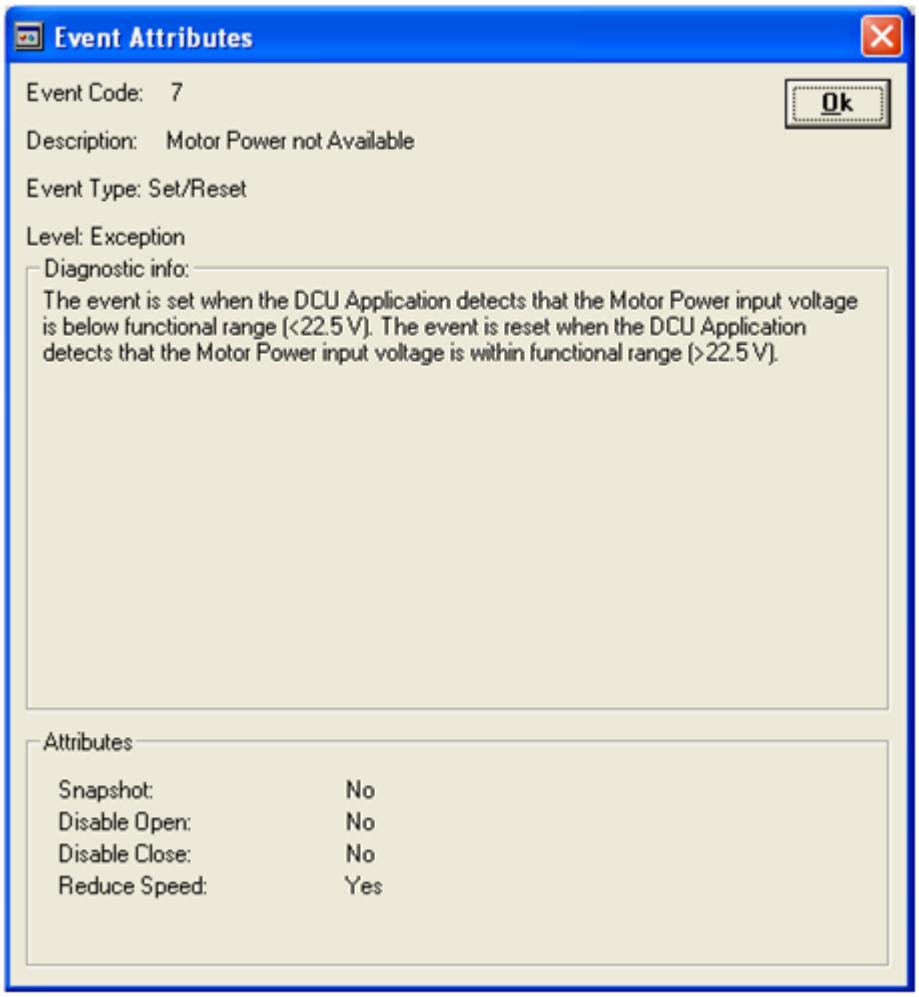


Figure 5-43: Event Attributes Window

The following information is presented in the Event Attributes Window:

- Event Code;
- Event Description;
- Event Type;
- Level;
- Diagnostic information;
- Attributes.

5.3.29.3 Related Information

None

5.3.29.4 Cautions and Warnings

The mere launching of the window will only display the EDF that is currently in the PTU Application database; the EDF needs to be downloaded from the DCU to display the most current file in the DCU (see Section 5.3.30).

5.3.30 Update EDF and OPDF

5.3.30.1 Scope

The following subsections provide information on how to download the Operating Parameter Description File and Event Description File from the DCU for updating the PTU Application's database.

5.3.30.2 Method

This function is found under the menu *Tools → Update EDF and OPDF*. A message window (Figure 5-44) pops up to show the download process when the PTU Application is performing the EDF&OPDF download from the DCU. The window is closed after the download is finished and the EDF (Figure 5-42) and OPDF (Figure 5-41) windows are displayed.

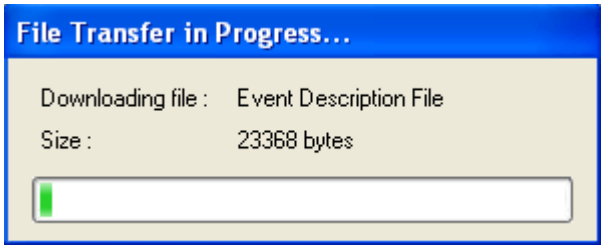


Figure 5-44: EDF Download Progress Window

The EDF and OPDF will be automatically updated whenever the EDF or OPDF version number requested by the PTU from the DCU is greater than the ones loaded in the PTU Application.

5.3.30.3 Related Information

None

5.3.30.4 Cautions and Warnings

Refer to file transfer related errors listed in Table 5-2 for possible errors that may occur during transfer of larger files, as is the case for this function.

5.3.31 Help Contents

5.3.31.1 Scope

The following subsections provide information on how to access the PTU Application help that is available within the application.

5.3.31.2 Method

To access the PTU help information, click *Help → Help Contents* from the PTU menu.

The help information provided is this PTU Application Software User Manual and it is opened with a document viewer.

5.3.31.3 Related Information

None

5.3.31.4 Cautions and Warnings

None

5.3.32 About Window

5.3.32.1 Scope

The following subsections provide information on how to view the version information for the PTU Application.

5.3.32.2 Method

Click *Help → About* from the PTU menu to open the PTU About window. As shown in Figure 5-45, the PTU user can find the PTU Application version and copyright information in the About window.

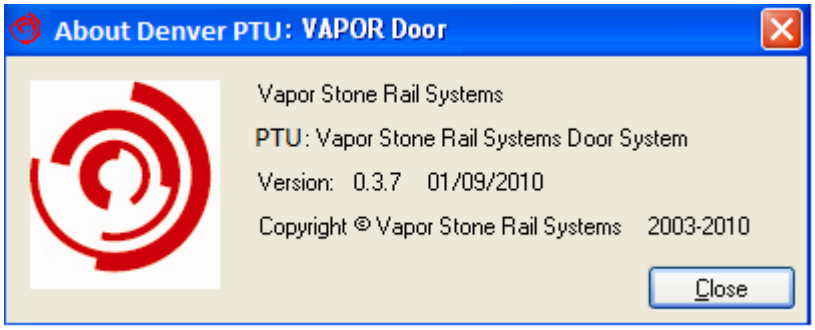


Figure 5-45: About Window

Note: The version numbers in these screenshots are for example purposes only and may not reflect the latest software version.

The About window provides a command button:

- Close button: Close the About window.

5.3.32.3 Related Information

None

5.3.32.4 Cautions and Warnings

None

5.3.33 Project File Save / Open

5.3.33.1 Scope

The following subsections provide information on how to save the project file and what it contains.

5.3.33.2 Method

The data saved in the project file includes:

- Location ID
- Car Number
- SCI Versions
- Event Log
- Snapshots
- Graphs
- System Counters

Saving the project file:

1. From the menu, click *File → Save Project As.*

2. A standard windows file save prompt will appear. Choose the destination folder and type the name of the file to be saved.

Opening a project file:

1. From the menu, click *File → Open*.

2. Select the *.VPP file to open

5.3.33.3 Related Information

- Information intended to be saved with the project file must first be downloaded from a DCU. This means that if the Event Log was not downloaded before saving, the project file will not contain the Event Log data.

- If the option “Save Project” is chosen, the PTU Application will replace the currently opened *.VPP file with the new contents. If no project file was opened, the “save as” window will appear.

5.3.33.4 Cautions and Warning

It is not recommended to open a project file while logged in to a DCU for fear of confusion.

If is not recommended to create a project file using Event Log data originating from a *.txt file loaded using the Event Log View “Load” button. The project file must be saved with information loaded from a direct link to a DCU.

5.3.34 Print

5.3.34.1 Scope

The following subsections provide information on how to print the DCU Event Log and the associated Snapshots information that is currently in the PTU Application's database; this function also offers the option of printing the PTU Project File Summary.

5.3.34.2 Method

From the menu, click *File → Print*. A window (see Figure 5-46) prompts the user to select the type of information to be printed.

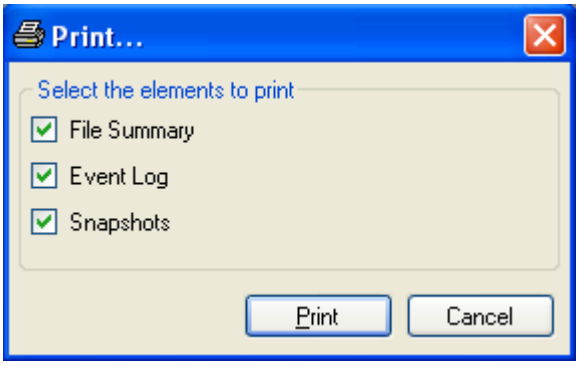


Figure 5-46: Print User Interface

The following information is presented in the window:

- File Summary checkbox: Select the PTU project File Summary to be printed;
- Event Log checkbox: Select the Event Log to be printed;
- Snapshot checkbox: Select the Snapshot to be printed.

The Print window also provides command buttons:

- Print button: Proceed with the printing of the selection;
- Cancel button: Cancel the printing action.

5.3.34.3 Related Information

None

5.3.34.4 Cautions and Warning

None

5.3.35 Communication Debug

5.3.35.1 Scope

The following subsections provide information on how to view the communication packets coming into and going out of the PTU Application.

5.3.35.2 Method

Click *Tools → Debug → Communication Debug* from the PTU Menu to view messages sent to and received from DCU.

The Communication Debug Window shows communication activities between the PTU Application and the DCU Application and could be used to troubleshoot PTU Application communication problems.
The PTU Status bar indicates the current Communication Device mode.

The following information is presented:

- [out]: Commands sent by the PTU to the DCU.
- [in]: Response from the DCU received by the PTU.

The Communication Debug window also provides a command button:

- New document icon: Clear the Communication Log.

5.3.35.3 Related Information

➢ For additional help in diagnosing communication related problems, the PTU Error Log should also be observed (see Section 5.3.36).

5.3.35.4 Cautions and Warnings

It should be noted that this feature is not for debugging or diagnosing the DCU, but is meant for debugging or diagnosing only the communications link between the PTU Application and the DCU.

5.3.36 Error Log

5.3.36.1 Scope

The following subsections provide information on how to view the Error Log related to the PTU Application.

5.3.36.2 Method

Click *Tools* → *Debug* → *Error Log* from the PTU Menu to view error messages.

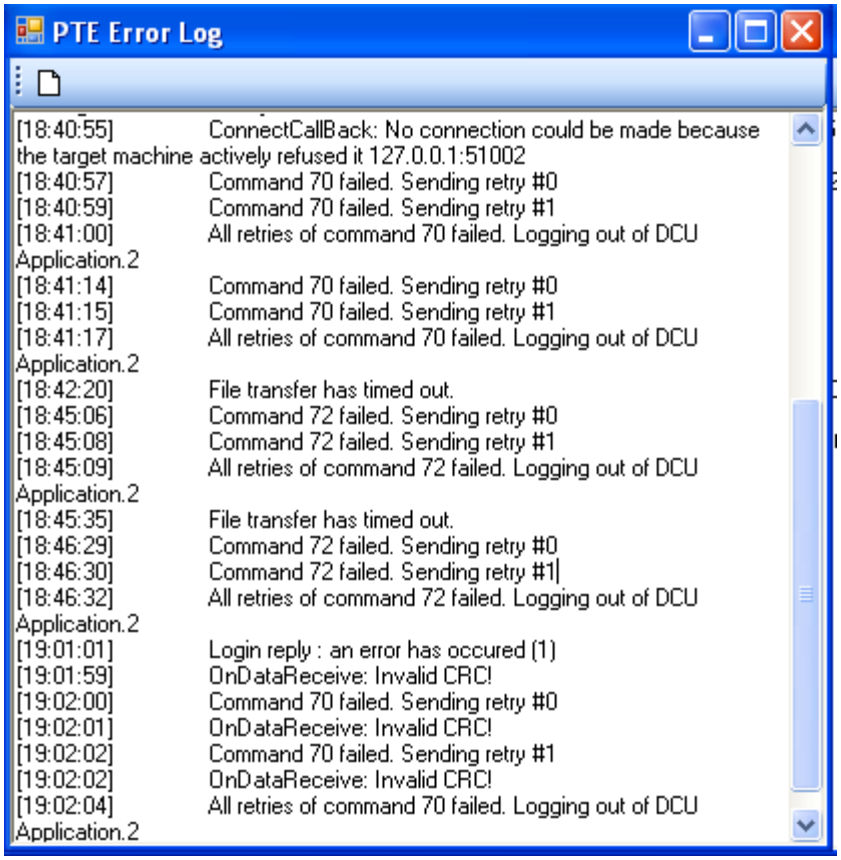


Figure 5-47: Error Log Window

The Error Log window shows internal PTU Application activities and could be used to troubleshoot PTU Application problems.

The Error Log window also provides a command button:

- New document icon: Clear the Error Log.

5.3.36.3 Related Information

- For additional help in diagnosing communication related problems, the PTU Communication Debug Log should also be observed (see Section 5.3.35).

- Portable Test Interface Control Document (ICD) provides information regarding the communication protocol used by the PTU.

5.3.36.4 Cautions and Warnings

It should be noted that this feature is not for debugging or diagnosing the DCU, but is meant for debugging or diagnosing only the PTU Application and its communications link with the DCU.

5.3.37 Error and Exception

5.3.37.1 Scope

The following subsections provide information on how certain errors encountered during use of the PTU Application are displayed to the user.

5.3.37.2 Method

When an error or exception occurs, a window pops up indicating the error.

As shown in Figure 5-48, the following information is presented in the window:

- Error Message: The description of error or warning;
- OK button: Close the Error message window and resume operation.

Figure 5-48: Error/Warning Message Window

5.3.37.3 Related Information

None

5.3.37.4 Cautions and Warnings

Refer to Table 5-2 for a complete listing of PTU Application errors.

5.4 Related Processing

There is no related batch, offline, or background processing performed by the software that is not invoked directly by the user and is not described in Section 5.3.

5.5 Data Backup

The data created by the PTU Application consists of Event Log text files, Memory Dump binary files, and project data files that have been saved by the user. All these data reside in the "\DENVERPTU\DATA\DOOR" directory of the drive the PTU Application is installed. To make a backup, simply copy the contents of this directory onto the backup media.

5.6 Recovery from Errors, Malfunctions, and Emergencies

For recovery from errors, see Section 5.7 for the appropriate corrective action.

In the event that a software malfunction occurs, restart the PTU computer system. If the PTU Application is still malfunctioning after the restart, uninstall and reinstall the PTU Application (see Section 4.1.3).

5.7 Messages

The following table lists all errors and warnings that can occur with the PTU Application.

Table 5-2: Error/Warning List

Name	Type	Description	Corrective action
An error has occurred during the file transfer. Please view the Error Log for more details.	Error	There was a problem with the file transfer.	• Open the Error Log for more information. • Retry the file transfer. If the error is still there, ask vendor.
DCU App. upgrade failure.	Error	There was a problem during the upgrade of the DCU Application.	• Check the connection between the PTU computer and the DCU unit. • Reboot the DCU unit. • Restart the PTU Application. • Redo the upgrade of DCU Application. If the error is still there, ask vendor.
DCU replied with a login error.	Warning	The DCU refuses the PTU login request.	• Reboot the DCU unit. • Retry the login request.
File does not exist. Verify that the correct file name was given.	Warning	The user tried to open a file that does not exist.	• Verify the path and filename. • Redo the open file with the correct file.
Opening the file failed. Verify that the file is a valid PTU Project File (.vpp).	Error	The user data file is not the correct format or the file is corrupted.	• Verify that the file is a .vpp file. • Redo the open file with an uncorrupted .vpp file.
The entered Password is incorrect	Warning	The Password entered did not match the required Password.	• Make sure Caps Lock is not on. • Make sure the password is correct. • Re-enter the Password.
Unable to save file. Check the error log for more details.	Error	There was a problem in saving the file.	• Open the Error Log for more information. • Check if the filename used for saving belongs to an existing read only file. • Check if there is enough disk space remaining. • Redo the file save with another filename.

5.8 Quick-Reference Guide

The PTU Control Menu described in Section 5.3.2 can be used as a quick-reference guide.

6 NOTES

6.1 Definitions

None

6.2 Acronyms

°F	Degree Fahrenheit
A	Ampere
API	Application Programming Interface
App	Application
ASCII	American Standard Code for Information Interchange
CDRL	Contract Data Requirements List
CD-ROM	Compact Disk Read Only Memory
COM	Communication
CPLD	Complex Programmable Logic Device
CRC	Cyclic Redundancy Check
CRC32	32-bit Cyclic Redundancy Check
DCU	Door Control Unit
ECD	Engineering Change Document
EDF	Event Description File
GB	Gigabytes
GHz	Gigahertz
h:m:s	Hour:Minute:Second
HD	Hard Disk
Hex	Hexadecimal
IBM	International Business Machines
ID	Identification
LED	Light-Emitting Diode
m-d-y	Month-Day-Year
MB	Megabytes
MDI	Multiple-Document Interface
MDS	Monitoring and Diagnostic System
MHz	Megahertz
ms	Millisecond
MS	Microsoft
MU	Multiple Unit
NVRAM	Non-Volatile Random Access Memory
OPDF	Operating Parameters Description File
P/N	Part Number
PC	Personal Computer
PSS	Panel Sensing Switch
PTU	Portable Test Unit
PWM	Pulse Width Modulation
RAM	Random Access Memory
Rx	Receive
s	Second
SCI	Software Configuration Item
SCMP	Software Configuration Management Plan
SDD	Software Design Description
SLS	Solenoid Lock Switch
SPMP	Software Project Management Plan
SPR	System Problem Report
SQAP	Software Quality Assurance Plan
SRAM	Static Random Access Memory
SRS	Software Requirements Specification
SFD	Software System Functional Description
std	Standard
SUM	Software User Manual
SVGA	Super Video Graphics Array
SVVP	Software Verification and Validation Plan
SW	Software
TL	Trainline
TS	Technical Specification
Tx	Transmit
V	Volt
VSRS	Vapor Stone Rail Systems
Win32	[Microsoft] Windows 32-bit
XP	[Microsoft Windows] eXPerience
XXX	Value unknown at the moment

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